

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent

12408494

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

Oh; Won Sik et al.

Display device with light emitting elements on parallel electrode branches extending from separate parallel electrodes

Abstract

A display device includes a first electrode stem and a second electrode stem extended in a first direction and spaced from each other, a first electrode branch branching off from the first electrode stem and extended in a second direction, a second electrode branch branching off from the second electrode stem and extended in the second direction, a third electrode between the first electrode branch and the second electrode branch and one or more light-emitting elements between the first electrode branch and the third electrode and between the third electrode and the second electrode branch, wherein the third electrode is extended in the second direction, and both ends of the third electrode in the second direction are spaced from the first electrode stem and from the second electrode stem, respectively.

Inventors: Oh; Won Sik (Hwaseong-si, KR), Kang; Sin Chul (Hwaseong-si, KR), Kim; Dae Hyun (Hwaseong-si, KR), Cho; Hyun Min (Seoul, KR)

Applicant: SAMSUNG DISPLAY CO., LTD. (Yongin-si, KR)

Family ID: 69164740

Assignee: Samsung Display Co., Ltd. (Yongin-si, KR)

Appl. No.: 18/669514

Filed: May 20, 2024

Prior Publication Data

Document Identifier

US 20240313165 A1

Publication Date

Sep. 19, 2024

Foreign Application Priority Data

KR

10-2018-0084255

Jul. 19, 2018

Related U.S. Application Data

division parent-doc US 17260944 US 11990566 WO PCT/KR2018/015859 20181213 child-doc US 18669514

Publication Classification

Int. Cl.: H10H20/831 (20250101); H10H20/857 (20250101); H10H29/14 (20250101)

U.S. Cl.:

CPC H10H20/8312 (20250101); H10H20/857 (20250101); H10H29/142 (20250101);

Field of Classification Search

CPC: H10H (29/142); H10H (20/831); H10H (20/8312); H01L (25/167); H10D (10/821)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8994620	12/2014	Toya	N/A	N/A
9590139	12/2016	Lai et al.	N/A	N/A
10211418	12/2018	Im et al.	N/A	N/A
10714701	12/2019	Im et al.	N/A	N/A
11127914	12/2020	Im et al.	N/A	N/A
2008/0251381	12/2007	Shibata et al.	N/A	N/A
2011/0089850	12/2010	Shibata et al.	N/A	N/A
2011/0254043	12/2010	Negishi et al.	N/A	N/A
2011/0316009	12/2010	Fukasawa	257/E33.062	H01L 25/0753
2014/0145237	12/2013	Do et al.	N/A	N/A
2015/0154915	12/2014	Toya	N/A	N/A
2017/0200866	12/2016	Wu et al.	N/A	N/A
2018/0019369	12/2017	Cho et al.	N/A	N/A
2018/0019426	12/2017	Im et al.	N/A	N/A
2018/0138157	12/2017	Im et al.	N/A	N/A
2018/0175009	12/2017	Kim et al.	N/A	N/A
2018/0175106	12/2017	Kim et al.	N/A	N/A
2018/0198018	12/2017	Kim et al.	N/A	N/A
2018/0211582	12/2017	Sakariya et al.	N/A	N/A
2019/0013298	12/2018	Lai	N/A	N/A
2019/0157512	12/2018	Jung	N/A	N/A
2019/0296200	12/2018	Im et al.	N/A	N/A
2020/0090574	12/2019	Sakariya et al.	N/A	N/A
2020/0203583	12/2019	Im et al.	N/A	N/A
2021/0193885	12/2020	Im et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107256877	12/2016	CN	N/A
107610602	12/2017	CN	N/A
107623013	12/2017	CN	N/A
2012-009622	12/2011	JP	N/A
2018-518711	12/2017	JP	N/A
10-2012-0122159	12/2011	KR	N/A
10-1436123	12/2013	KR	N/A
10-2015-0006798	12/2014	KR	N/A
10-2016-0059574	12/2015	KR	N/A
10-2018-0007376	12/2017	KR	N/A
10-2018-0009015	12/2017	KR	N/A
10-2018-0009116	12/2017	KR	N/A
10-2018-0033927	12/2017	KR	N/A
2018-0055021	12/2017	KR	N/A
10-2018-0059801	12/2017	KR	N/A
10-2018-0072909	12/2017	KR	N/A
201721617	12/2016	TW	N/A

OTHER PUBLICATIONS

Chinese Office Action dated Sep. 30, 2024 regarding Chinese Patent Application No. 201880095677.4 corresponding to U.S. Appl. No. 18/669,514 (6 pages). cited by applicant
Chinese Office Action dated Jan. 18, 2024 regarding Chinese Patent Application No. 201880095677.4 corresponding to U.S. Appl. No. 17/260,944 (9 pages). cited by applicant
International Search Report for corresponding International Application No. PCT/KR2018/015859, dated Apr. 18, 2019, 4pp. cited by applicant

Primary Examiner: Haider; Wasiul

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/260,944, filed Jan. 15, 2021, which is a National Phase Patent Application and claims priority to and the benefit of International Application Number PCT/KR2018/015859, filed on Dec. 13, 2018, which claims priority to Korean Patent Application Number 10-2018-0084255, filed on Jul. 19, 2018, the entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a display device, and more particularly, to a display device including an array of inorganic light-emitting diodes.

BACKGROUND ART

(2) Display devices become more and more important as multimedia technology evolves. Accordingly, a variety of types of display devices such as organic light-emitting display (OLED)

devices and liquid-crystal display (LCD) devices are currently used.

(3) Display devices are for displaying images and include a display panel such as an organic light-emitting display panel or a liquid-crystal display panel. Among them, light-emitting display panels may include light-emitting elements. For example, light-emitting diodes (LEDs) may include an organic light-emitting diode (OLED) using an organic material as a fluorescent material, and an inorganic light-emitting diode using an inorganic material as a fluorescent material.

(4) An organic light-emitting diode (OLED) uses an organic material as the fluorescent material of the light-emitting elements. It is desirable that the fabricating process is simple and the element has flexibility. However, it is known that the organic materials are vulnerable to high-temperature driving environments and the efficiency of blue light is relatively low.

(5) In contrast, an inorganic light-emitting diode uses an inorganic semiconductor as the fluorescent material and is desirable that it has durability in high-temperature environments and that it has a higher efficiency of blue light than organic light-emitting diodes. Previously, the process of fabricating inorganic light-emitting diodes had its shortcomings. However, a transfer method using dielectrophoresis (DEP) has been developed, and thus the shortcoming has been overcome. Therefore, researches on inorganic light-emitting diodes are ongoing because they have better durability and efficiency compared to the organic light-emitting diodes.

DISCLOSURE

Technical Problem

(6) Aspects of the present disclosure provide a display device in which light-emitting elements are connected in series, so that the capacity of driving transistors can be reduced, and the voltage-dividing efficiency and the power loss due to the line resistance can be improved.

(7) Additional advantages, subjects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or may be learned from practice of the invention.

Technical Solution

(8) According to an aspect of the present disclosure, a display device includes a first electrode stem and a second electrode stem extended in a first direction and spaced from each other, a first electrode branch branching off from the first electrode stem and extended in a second direction, a second electrode branch branching off from the second electrode stem and extended in the second direction, a third electrode located between the first electrode branch and the second electrode branch and one or more light-emitting elements located between the first electrode branch and the third electrode and between the third electrode and the second electrode branch, wherein the third electrode is extended in the second direction, and both ends of the third electrode in the second direction are spaced from the first electrode stem and from the second electrode stem, respectively.

(9) The first electrode branch may be extended opposite to the second electrode branch in the second direction.

(10) The light-emitting elements may include a first light-emitting element located between the first electrode branch and the third electrode and a second light-emitting element located between the third electrode and the second electrode branch, wherein the first light-emitting element and the second light-emitting element may be connected in series between the first electrode branch and the second electrode branch.

(11) A first end of the first light-emitting element may be connected to the first electrode branch and a second end of the first light-emitting element may be connected to the third electrode, wherein the first end or the second end of the first light-emitting element may include a p-type semiconductor layer and the second end or the first end of the first light-emitting element may include an n-type semiconductor layer, and wherein the first light-emitting element may include an active layer between the p-type semiconductor layer and the n-type semiconductor layer.

(12) The display device may include a plurality of first electrode branches branching off from the first electrode branch, and the second electrode branch may be between the plurality of first

electrode branches.

(13) The third electrode may be located at least between the second electrode branch and the first electrode branch located on one side of the second electrode branch, and between the second electrode branch and the first electrode branch located on an opposite side of the second electrode branch.

(14) The display device may include two or more third electrodes, and wherein the light-emitting elements may be between adjacent ones of the third electrodes.

(15) The light-emitting elements may include a third light-emitting element electrically connected on the one side of the second electrode branch and a fourth light-emitting element electrically connected on the opposite side of the second electrode branch, wherein the third light-emitting element and the fourth light-emitting element may be connected in parallel between the one side and the opposite side of the second electrode branch.

(16) The plurality of first electrode branches and the two or more third electrodes may be symmetrically arranged with respect to the second electrode branch.

(17) The third electrode may include a third electrode fragment protruding from the first electrode stem and the second electrode stem in the second direction and a third electrode branch spaced from the third electrode fragment, the third electrode branch extending from the third electrode fragment.

(18) The third electrode may include two or more third electrode branches spaced from each other between the first electrode branch and the second electrode branch, and wherein the light-emitting elements may include a fifth light-emitting element having an end electrically connected to the first electrode stem and another end connected to the third electrode branch, a sixth light-emitting element having both ends electrically connected to the two or more third electrode branches and a seventh light-emitting element having one end electrically connected to the third electrode branch and another end connected to the second electrode branch, wherein the fifth light-emitting element, the sixth light-emitting element and the seventh light-emitting element may be connected in series between the first electrode branch and the second electrode branch.

(19) According to another aspect of the present disclosure, a display device includes a first electrode stem and a second electrode line extended in a first direction and spaced from each other, a first electrode branch branching off from the first electrode stem and extended in a second direction, a second electrode branch spaced from the first electrode branch and extended in the second direction, the second electrode branch being terminated while spaced from the first electrode stem, a third electrode located between the first electrode branch and the second electrode branch and extended in the second direction, both ends of the third electrode being terminated while spaced from the first electrode stem and the second electrode line and one or more light-emitting elements located among the first electrode branch, the second electrode branch and the third electrode, wherein the second electrode branch is electrically connected to the second electrode line.

(20) The first electrode stem and the second electrode line may be located at different line layers, and wherein the second electrode branch may be electrically connected to the second electrode line through a contact hole.

(21) The first electrode stem may further include a second electrode fragment protruding from the first electrode stem in the second direction.

(22) The second electrode fragment may be extended in the second direction and is aligned with the second electrode branch.

(23) The first electrode stem may be electrically connected to a thin-film transistor, wherein the second electrode line is electrically connected to a common power supply line, wherein the light-emitting elements may include a first light-emitting element located between the first electrode branch and the third electrode, and a second light-emitting element located between the third electrode and the second electrode branch, and wherein the first light-emitting element and the

second light-emitting element may be connected in series between the first electrode branch and the second electrode branch.

(24) According to the other aspect of the present disclosure, a display device includes a first electrode connected to a thin-film transistor, a second electrode connected to a common power supply line, a floating electrode between the first electrode and the second electrode, a first light-emitting diode having one end electrically connected to the first electrode and another end electrically connected to the floating electrode and a second light-emitting diode having one end electrically connected to the floating electrode and another end electrically connected to the second electrode, wherein the first light-emitting diode and the second light-emitting diode are connected in series between the first electrode and the second electrode.

(25) The first electrode, the second electrode and the floating electrode may be extended in a first direction, and wherein the first light-emitting diode and the second light-emitting diode in series may be extended in a second direction.

(26) The first light-emitting diode and the second light-emitting diode each may include a first conductivity-type semiconductor layer, an active layer, and a second conductivity-type semiconductor layer, wherein one of the first conductivity-type semiconductor layer and the second conductivity-type semiconductor layer may be of p-type and other one may be of n-type, and wherein the first conductivity-type semiconductor layer of the first light-emitting diode may be electrically connected to the first electrode, and the second conductivity-type semiconductor layer of the first light-emitting diode may be electrically connected to the floating electrode, the first conductivity-type semiconductor layer of the second light-emitting diode may be electrically connected to the floating electrode, and the second conductivity-type semiconductor layer of the second light-emitting diode may be electrically connected to the second electrode.

(27) The length of the first light-emitting diode and the second light-emitting diode may range from 3 μm to 6 μm .

(28) The details of other embodiments are included in the detailed description and the drawings.

(29) According to an exemplary embodiment of the present disclosure, a display device includes a floating electrode between a first electrode and a second electrode, so that light-emitting elements electrically connected between the electrodes can be connected in series between the first electrode and the second electrode. Accordingly, during the process of fabricating display devices, it is easier to design the capacity of driving thin-film transistors, and it is possible to improve voltage-dividing efficiency and power loss due to line resistance.

(30) Advantageous effects according to the present disclosure are not limited to those mentioned above, and various other advantageous effects are included herein.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a plan view of a display device according to an exemplary embodiment of the present disclosure.

(2) FIG. 2 is an equivalent circuit diagram of a single pixel of the display device of FIG. 1.

(3) FIG. 3 is an equivalent circuit diagram of a pixel of a display device according to another exemplary embodiment of the present disclosure.

(4) FIG. 4 is a cross-sectional view taken along the line I-I' of FIG. 1.

(5) FIG. 5 is a plan view of a display device according to another exemplary embodiment of the present disclosure;

(6) FIG. 6 is a cross-sectional view taken along the lines II-II' and III-III' of FIG. 5.

(7) FIG. 7 is a view showing the light-emitting element according to an exemplary embodiment of the present disclosure.

- (8) FIGS. **8** to **10** are plan views schematically showing a method of fabricating a display device according to an exemplary embodiment of the present disclosure.
- (9) FIGS. **11** to **13** are plan views of a display device according to another exemplary embodiment of the present disclosure.
- (10) FIG. **14** is a plan view of a display device according to yet another exemplary embodiment of the present disclosure.
- (11) FIGS. **15** to **17** are plan views for illustrating a method of fabricating the display device of FIG. **14**.
- (12) FIGS. **18** and **19** are plan views of a display device according to another exemplary embodiment of the present disclosure.
- (13) FIG. **20** is a plan view of a display device according to yet another exemplary embodiment of the present disclosure.
- (14) FIGS. **21** to **23** are plan views for illustrating a method of fabricating the display device of FIG. **20**.

MODES OF THE INVENTION

- (15) The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.
- (16) It will also be understood that when a layer is referred to as being “on” another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. The same reference numbers indicate the same components throughout the specification.
- (17) It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed below could be termed a second element without departing from the teachings of the present invention. Similarly, the second element could also be termed the first element.
- (18) Hereinafter, exemplary embodiments of the present disclosure will be described with reference to the accompanying drawings.
- (19) FIG. **1** is a plan view of a display device according to an exemplary embodiment of the present disclosure.
- (20) The display device **10** may include one or more regions defined as pixels PX (e.g., one of pixels PX1, PX2, PX3). The plurality of pixels PX may be disposed on a display part of the display device **10** so that light of a particular wavelength band can exit the display device **10**. Although three pixels PX1, PX2 and PX3 are depicted in FIG. **1** by way of example, it is to be understood that the display device **10** may include more pixels. Although the cross-sectional view of FIG. **1** shows the pixels PX arranged only along one direction, i.e., a first direction D1, the pixels PX may be arranged along a second direction D2 intersecting with the first direction D1 as well. In addition, the pixels PX shown in FIG. **1** may be divided into a plurality of sub-pixels, each of which may form a pixel PX (e.g., one of pixels PX1, PX2, PX3). The pixels may not be necessarily arranged along parallel as shown in FIG. **1** but may be arranged in a variety of ways, e.g., in the vertical direction (or the second direction), in a zigzag pattern, etc.
- (21) Although not shown in the drawings, the display device **10** may include an emission portion in which the light-emitting elements **350** are disposed to emit light of a color, and a non-emission portion that is defined as the region other than the emission portion. The non-emission portion may be covered by some members so that it is not visible from outside the display device **10**. In the non-emission portion, a variety of elements for driving the light-emitting elements **350** disposed in the emission portion are located. For example, in the non-emission portion, lines for applying electrical

signals to the emission portion, circuits, drivers, etc., may be disposed. It is, however, to be understood that the present disclosure is not limited thereto.

(22) Each of the plurality of pixels PX may include one or more light-emitting elements **350** that emit light of particular wavelength bands to represent colors. The light emitted from the light-emitting elements **350** can be seen outside the display device **10** via the emission portion. In an exemplary embodiment, pixels PX that represent different colors may include light-emitting elements **350** that emit different colors. For example, a first pixel PX1 for representing red may include a light-emitting element **350** for emitting red light, a second pixel PX2 for representing green may include a light-emitting element **350** for emitting green light, and a third pixel PX3 for representing blue may include a light-emitting element **350** for emitting blue light. It is, however, to be understood that the present disclosure is not limited thereto. In some implementations, the pixels representing different colors may include the light-emitting elements **350** for emitting the same color (e.g., blue), and a wavelength conversion layer or a color filter may be located on in the emission path to represent colors of the pixels. It is, however, to be understood that the present disclosure is not limited thereto. In some implementations, adjacent pixels PX may emit light of the same color.

(23) Referring to FIG. 1, the display device **10** may include a plurality of electrodes **310**, **320** and **330** and a plurality of light-emitting elements **350**. At least some of the electrodes **310**, **320** and **330** may be disposed in each of the pixels PX and may be electrically connected to the light-emitting elements **350** to apply electrical signals so that the light-emitting elements **350** can represent a particular color.

(24) In addition, at least some of the electrodes **310**, **320** and **330** may be utilized to form an electric field within the pixel PX to align the light-emitting elements **350**. Specifically, when the light-emitting elements **350** that emit different colors are aligned in the plurality of pixels PX, it is necessary to accurately align the different light-emitting elements **350** for each of the pixels PX. The light-emitting elements **350** may be aligned using the dielectrophoresis as follows: a solution containing the light-emitting elements **350** is applied onto the display device **10**, and an AC electric power is applied to it to form an electric field, and a dielectrophoresis force is applied to the light-emitting elements **350** to align them.

(25) The plurality of electrodes **310**, **320** and **330** may include a first electrode **310**, a second electrode **320**, and a third electrode **330**. In an exemplary embodiment, the first electrode **310** may be disconnected from pixel to pixel PX, while the second electrode **320** may be a common electrode connected across the plurality of pixels PX. The first electrode **310** may be an anode electrode of the light-emitting elements **350**, while the second electrode **320** may be a cathode electrode of the light-emitting elements **350**. It is, however, to be understood that the present disclosure is not limited thereto. That is to say, the first electrode **310** may be a cathode electrode of the light-emitting elements **350**, while the second electrode **320** may be an anode electrode of the light-emitting elements **350**.

(26) The first electrode **310** and the second electrode **320** may include electrode stems **310S** and **320S** extended in the first direction D1, respectively, and one or more electrode branches **310B** and **320B** branching off from the electrode stems **310S** and **320S**, respectively, and extended in a second direction D2 intersecting the first direction D1.

(27) Specifically, the first electrode **310** may include a first electrode stem **310S** extended in the first direction D1, and at least one first electrode branch **310B** branching off from the electrode stem **310S**, and extended in the second direction D2. Although not shown in the drawings, one end of the first electrode stem **310S** may be connected to a signal application pad, the other end thereof may be extended in the first direction D1 and electrically disconnected from pixel to pixel PX. The signal application pad may be connected to the display device **10** or an external power source to apply an electrical signal to the first electrode stem **310S** or to apply an AC power to align the light-emitting elements **350**.

(28) A first electrode stem **310S** of a pixel may be substantially in line with a first electrode stem **310S** of a pixel adjacent to the pixel in the same row (e.g., in the first direction **D1**). In other words, both ends of the first electrode stem **310S** of a pixel are terminated between the pixel and an adjacent pixel **PX** and separated from it, and the first electrode stem **310S** of the adjacent pixel may be in line with the first electrode stem **310S** of the pixel. The first electrode stem **310S** may be formed as a single stem electrode during the fabrication process, and then may be cut using a laser or the like after the light-emitting elements **350** are aligned. Accordingly, the first electrode stem **310S** disposed in each of the pixels **PX** may apply different electrical signals to different electrode branches **310B**, such that the first electrode branches **310B** may be driven individually.

(29) The first electrode branch **310B** may branch off from at least a portion of the first electrode stem **310S** to be extended in the second direction **D2** and may be terminated while being spaced apart from the second electrode stem **320S** that is located opposite to the first electrode stem **310S**. That is to say, in a pixel **PX**, an end of the first electrode branch **310B** may be connected to the first electrode stem **310S**, and the other end of the first electrode branch **310B** may be spaced apart from the second electrode stem **320S**. Because the first electrode branch **310B** is connected to the first electrode stem **310S** that is electrically separated from pixel to pixel, different pixels **PX** can receive different electrical signals.

(30) In addition, the first electrode branch **310B** may be spaced apart from the second electrode branch **320B** and the third electrode **330** to be described later. Although only one first electrode branch **310B** is shown in FIG. 1, more than one first electrode branches **310B** may be provided. In such a case, some of the first electrode branches **310B** may be electrically separated from the first electrode stem **310S** to form the third electrode **330** to be described later. A more detailed description thereon will be given below.

(31) The second electrode **320** may include a second electrode stem **320S** extended in the first direction **D1** and spaced apart from and located opposite to the first electrode stem **310S**, at least one second electrode branch **320B** branching off from the second electrode stem **320S** and extended in the second direction **D2** and spaced apart from and facing the first electrode branch **310B**. Like the first electrode stem **310S**, an end of the second electrode stem **320S** may be connected to the signal application pad. It is to be noted that the other end of the second electrode stem **320S** may be extended across the plurality of pixels **PX** adjacent to one another in the first direction **D1**. That is to say, the second electrode stem **320S** may be electrically connected between the pixels **PX**. Accordingly, both ends of the second electrode stem **320S** in a pixel are connected to ends of adjacent pixels on either sides, so that the same electrical signal can be applied to the pixels.

(32) The second electrode branch **320B** may branch off from at least a portion of the second electrode stem **320S** to be extended in the second direction **D2** and may be terminated while being spaced apart from the first electrode stem **310S**. That is to say, in a pixel **PX**, an end of the second electrode branch **320B** may be connected to the second electrode stem **320S**, and the other end of the second electrode branch **320B** may be spaced apart from the first electrode stem **310S**. Because the second electrode branch **320B** is connected to the second electrode stem **320S** that is electrically connected from pixel to pixel, different pixels **PX** can receive the same electrical signal.

(33) In addition, the second electrode branch **320B** may be spaced apart from the first electrode branch **310B** and the third electrode **330**. Because the first electrode stem **310S** and the second electrode stem **320S** are spaced apart from and located opposite to each other on opposite sides of the center of each pixel **PX**, the first electrode branch **310B** and the second electrode branch **320B** may be extended in opposite directions. In other words, the first electrode branch **310B** may be extended toward a side in the second direction **D2** and the second electrode branch **320B** may be extended toward the opposite side in the second direction **D2**, such that they may be disposed on the opposite sides with respect to the center of the pixel **PX**. It is, however, to be understood that the present disclosure is not limited thereto. The first electrode stem **310S** and the second electrode

stem **320S** may be disposed on the same side of the center of the pixel **PX** and may be spaced apart from each other. In such a case, the first electrode branch **310B** and the second electrode branch **320B** branching off from the electrode stems **310S** and **320S**, respectively, may be extended in the same direction.

(34) Although only one first electrode branch **320B** is shown in FIG. 1, more than one first electrode branches **310B** may be provided. In such a case, like the first electrode branches **310B**, some of the second electrode branches **320B** may be electrically separated from the second electrode stem **320S** to form the third electrode **330** to be described later.

(35) At least one third electrode **330** may be disposed between the first electrode branch **310B** and the second electrode branch **320B**. The third electrode **330** may have substantially the same shape as the first electrode branch **310B** or the second electrode branch **320B** and face them. The third electrode **330** may be extended in the second direction **D2**, and both ends thereof may be terminated such that they are spaced apart from the first electrode stem **310S** and the second electrode stem **320S** in the second direction **D2**. That is to say, the third electrode **330** may be electrically separated from the electrode stems **310S** and **320S**. In an exemplary embodiment, the third electrode **330** may be a floating electrode to which no electrical signal is directly applied from the electrode stems **310S** and **320S**. It is, however, to be understood that the present disclosure is not limited thereto.

(36) The third electrode **330** may be equally spaced apart from the first electrode branch **310B** and the second electrode branch **320B**. That is to say, the third electrode **330** may have a constant width between the first electrode branch **310B** and the second electrode branch **320B**. The light-emitting elements **350** described later are arranged between the first electrode branch **310B** and the third electrode **330** and between the third electrode **330** and the second electrode branch **320B**. The third electrode **330** may be spaced apart at a constant spacing so that the light-emitting elements **350** may be connected to the first electrode branch **310B** or the second electrode branch **320B** and the third electrode **330**. It is, however, to be understood that the present disclosure is not limited thereto. When the length of the third electrode **330** measured in the first direction **D1** is sufficiently long, the third electrode **330** may not necessarily equally be spaced apart from the first electrode branch **310B** and the second electrode **330**.

(37) In the drawing, one end of the third electrode **330** is aligned with one end of the first electrode branch **310B** and the other end of the third electrode **330** is aligned with one end of the second electrode branch **320B**. For example, when a plurality of third electrodes **330** is provided, the third electrodes **330** may be substantially aligned with the first electrode branch **310B** and the second electrode branch **320B** in the first direction **D1**. It is, however, to be understood that the present disclosure is not limited thereto. In some exemplary embodiments, at least one of both ends of the third electrode **330** may protrude from the end of the first electrode branch **310B** and the second electrode branch **320B**. In such a case, the plurality of third electrodes **330** may be arranged between the first electrode branch **310B** and the second electrode branch **320B**, and both ends of some of the third electrodes **330** may not be aligned with the ends but may be protruding from them.

(38) Such third electrodes **330** may be formed separately from the first electrode branch **310B** and the second electrode branch **320B**, but the present disclosure is not limited thereto. In some exemplary embodiments, the third electrode **330** may be formed to be substantially the same as the first electrode branch **310B** or the second electrode branch **320B**, and then may be separated during a subsequent disconnection process, such that they may be separated from the first electrode branch **310B** and the second electrode **320B**. Specifically, some of the third electrodes **330** may be electrically connected to the first electrode stem **310S** or the second electrode stem **320S**, and then may be disconnected by a laser or the like. A more detailed description thereon will be given below.

(39) The plurality of light-emitting elements **350** may be arranged between the first electrode branch **310B** and the third electrode **330** and between the third electrode **330** and the second

electrode branch **320B**. Specifically, some of the plurality of light-emitting elements **350** may have an end electrically connected to the first electrode branch **310B** and the other end electrically connected to the side of the third electrode **330** that faces the first electrode branch **310B**. In addition, some of the plurality of light-emitting elements **350** may have an end electrically connected to the other side of the third electrode **330** that faces the second electrode branch **310B** and the other end electrically connected to the second electrode branch **320B**.

(40) Contact electrodes **360** may be disposed on the first electrode branch **310B**, the second electrode branch **320B** and the third electrode **330** connected to the light-emitting elements **350**. The contact electrodes **360** may be in contact with the light-emitting elements **350** so that the light-emitting elements **350** may be electrically connected to the electrode branches **310B** and **320B** or the third electrode **330**. The contact electrodes **360** may come in contact with at least a side portion on both ends of each of the light-emitting elements **350**. Accordingly, the light-emitting elements **350** may each receive an electrical signal to emit light of a particular color.

(41) In some exemplary embodiments, one end of each of the light-emitting elements **350** in contact with the first electrode branch **310B** or the other side portion of the third electrode **330** may be an n-type or p-type doped conductive material layer, while the other end of each of the light-emitting elements **350** in contact with the one side of the third electrode **330** or the second electrode branch **320_B** may be a p-type or n-type doped conductive material layer. It is, however, to be understood that the present disclosure is not limited thereto.

(42) The plurality of light-emitting elements **350** may be arranged in parallel with one another. The light-emitting elements **350** may be equally spaced apart from one another. It is, however, to be understood that the present disclosure is not limited thereto. In some implementations, some of the light-emitting elements **350** may be disposed close to each other to form a group, and some other of the light-emitting elements **350** may be disposed close to each other to form another group that is spaced apart from the group. Alternatively, the light-emitting elements **350** may be arranged such that they are orientated in a direction with ununiform densities.

(43) As described above, the plurality of light-emitting elements **350** may be arranged between the first electrode branch **310B** and the third electrode **330** and between the third electrode **330** and the second electrode branch **320B** so that they may be electrically connected to one another. The distance between the first electrode branch **310B** and the second electrode branch **320B** may be larger than the length of the longer axis of the light-emitting elements **350**, whereas the distance between the third electrode **330** and the first electrode branch **310B** and the distance between the third electrode **330** and the second electrode branch **320B** may be smaller than the length of the longer axis of the light-emitting elements **350**. It is, however, to be understood that the present disclosure is not limited thereto.

(44) A light-emitting element **350** disposed between the third electrode **330** and the first electrode branch **310B** may be electrically connected to a light-emitting element **350** between the third electrode **330** and the second electrode branch **320B** in series. The light-emitting elements **350** may be arranged group by group in a pixel PX, so that they may form a series connection. Because the voltage applied to the light-emitting elements **350** is greater when the light-emitting elements **350** are connected in parallel, the voltage dividing efficiency can be increased. In addition, because the amount of current required to drive the light-emitting elements **350** connected in series decreases, the capacity of the driving transistor for applying the current to the light-emitting elements **350** can be reduced, and power loss due to the power supply lines can be reduced. As a result, the voltage efficiency and the luminous efficiency can be improved in driving the display device **10**.

(45) As shown in FIG. 1, the first electrode stem **310S** and the second electrode stem **320S** may be electrically connected to a thin-film transistor **120** or to a power supply line **161** (e.g., as shown in FIG. 4) to be described below through contact holes, e.g., a first contact hole CNTD and a second contact hole CNTS, respectively. Although the contact holes on the first electrode stem **310S** and the second electrode stem **320S** are formed in each of the pixels PX, the present disclosure is not

limited thereto. Because the second electrode stem **320S** may be extended and electrically connected to adjacent pixels PX as described above, in some exemplary embodiments, the second electrode stem **320S** may be electrically connected to a thin-film transistor through a single contact hole.

(46) FIG. 2 is an equivalent circuit diagram of a single pixel of the display device of FIG. 1. Although not shown in FIG. 2, the display device **10** may include a plurality of pixels PX. FIG. 2 shows a pixel at the *i*th row and *j*th column PX (*i*, *j*) as an example.

(47) Referring to FIG. 2, the pixel PX (*i*, *j*) of the display device **10** may include an *i*th scan line SL_{*i*}, a *j*th data line DL_{*j*}, a first switching element TR₁, a second switching element TR₂, light-emitting elements **350**, and a storage capacitor C_{st}.

(48) The first switching element TR₁ may be electrically connected to the *i*th scan line SL_{*i*}, the *j*th data line DL_{*j*}, and the second switching element TR₂. In an exemplary embodiment, the first switching element TR₁ and the second switching element TR₂ may be three terminal elements such as a thin-film transistor. In the following description, it is assumed that the first switching element TR₁ and the second switching element TR₂ are thin-film transistors.

(49) The first switching element TR₁ may include a control electrode electrically connected to the first scan line SL₁, an electrode electrically connected to the *j*th data line DL_{*j*}, and an other electrode electrically connected to the control electrode of the second switching element TR₂.

(50) The second switching element TR₂ may include a control electrode electrically connected to the other electrode of the first switching element TR₁, an electrode electrically connected to a first power supply line VDDL via which a first driving voltage VDD is applied, and an other electrode electrically connected to the light-emitting elements **350**.

(51) One electrode of the storage capacitor C_{st} may be electrically connected to the other electrode of the first switching element TR₁, and an other electrode of the storage capacitor C_{st} may be electrically connected to a first power supply line VDDL via which the first driving voltage VDD is supplied.

(52) The first switching element TR₁ may be turned on in response to a scan signal S_{*i*} supplied from the *i*th scan line SL_{*i*} and may provide a data signal D_{*j*} supplied from the *j*th data line DL_{*j*} to the storage capacitor C_{st}. The storage capacitor C_{st} may be charged with a voltage equal to a difference between the voltage of the received data signal D_{*j*} and the first driving voltage VDD. The second switching element TR₂ may control the amount of driving current supplied to the light-emitting elements **350** according to the voltage charged in the storage capacitor C_{st}. That is to say, the first switching element TR₁ may be a switching transistor, and the second switching element TR₂ may be a driving transistor.

(53) One end of the light-emitting elements **350** connected to the first electrode **310** may be electrically connected to the other electrode of the second switching element TR₂. The light-emitting elements **350** may receive a current through the other electrode of the second switching element TR₂. The other end of the light-emitting elements **350** connected to the second electrode **320** may be electrically connected to the second power supply line VSSL so that the second supply voltage VSS may be applied. The level of the first driving voltage VDD may be higher than the level of the second supply voltage VSS.

(54) As shown in FIG. 2, the light-emitting elements **350** may be connected in series between the second switching element TR₂ and the second power supply line VSSL via which the second supply voltage VESSEL is applied. As shown in FIG. 2, the number of the light-emitting elements **350** connected in series is not particularly limited. It is to be understood that a larger number of the light-emitting elements **350** may be connected in series. Accordingly, the first driving voltage VDD and the second supply voltage VSS for driving the light-emitting elements **350** can be more efficiently divided among the light-emitting elements **350**. In addition, the current flowing through the second switching element TR₂ decreases as the light-emitting elements **350** are connected in series, and thus it is easier to design the capacity of the thin-film transistors. Because the magnitude

of current decreases, the power loss on the power supply line or the first power supply line VDDL and the second power supply line VSSL can be reduced.

(55) In the example shown in FIG. 2, the display device **10** includes two switching elements, i.e., the first switching element TR1 and the second switching element TR2, and a single capacitor, i.e., the storage capacitor Cst. It is, however, to be understood that the present disclosure is not limited thereto. In some exemplary embodiments, the display device **10** may include a greater number of switching elements. Hereinafter, an equivalent circuit diagram of the display device **10** according to another exemplary embodiment will be described with reference to FIG. 3.

(56) FIG. 3 is an equivalent circuit diagram of a pixel of a display device according to another exemplary embodiment of the present disclosure. FIG. 3 shows a pixel at the *i*th row and the *j*th column PX (*i*, *j*) as an example.

(57) Referring to FIG. 3, the pixel PX (*i*, *j*) of the display device **10** may include an *i*th scan line SL_{*i*}, an (*i*−1)th scan line SL(*i*−1), a *j*th data line DL_{*j*}, an *i*th emission control line E_{*i*}, first to seventh switching elements TR1 to TR7, light-emitting elements **350**, and a storage capacitor Cst.

(58) One end of the light-emitting elements **350** connected to the first electrode **310** may be electrically connected to one electrode of the fifth switching element TR5 and to one electrode of the seventh switching element TR7. The light-emitting elements **350** may receive a current through the electrode of the fifth switching element TR5 and the electrode of the seventh switching element TR7. The other end of the light-emitting elements **350** connected to the second electrode **320** may be electrically connected to the second power supply line VSSL so that the second supply voltage VSS may be applied.

(59) One electrode of the first switching element TR1 is electrically connected to a first node N1, and an other electrode of the first switching element TR1 is electrically connected to a second node N2. A gate electrode of the first switching element TR1 is electrically connected to one end of the storage capacitor Cst through a third node N3. The first switching element TR1 is electrically connected to one end of the light-emitting elements **350** via the fifth switching element TR5. The first switching element TR1 may receive the data signal of the *j*th data line DL_{*j*} and supply a current to the light-emitting elements **350** according to the switching operation of the second switching element TR2.

(60) One electrode of the second switching element TR2 is electrically connected to the *j*th data line DL_{*j*}, and an other electrode of the second switching element TR2 is electrically connected to the first node N1. A gate electrode of the second switching element TR2 is electrically connected to the *i*th scan line SL_{*i*}. The other electrode of the second switching element TR2 is electrically connected to the first node N1, which is electrically connected to the first power supply line VDDL through the fourth switching element TR4. The second switching element may be turned on in response to a scan signal on the *i*th scan line SL_{*i*}, and may perform a switching operation to transfer the data signal transferred to the *j*th data line DL_{*j*} to the first switching element TR1.

(61) One electrode of the third switching element TR3 is electrically connected to the second node N2, and an other electrode of the third switching element TR3 is electrically connected to the third node N3. A gate electrode of the third switching element TR3 is electrically connected to the *i*th scan line SL_{*i*}. When a scan signal is applied to the *i*th scan line SL_{*i*}, the third switching element TR3 is turned on and the first switching element TR1 is diode-connected.

(62) One electrode of the fourth switching element TR4 is electrically connected to the first power supply line VDDL to receive the first driving voltage VDD. An other electrode of the fourth switching element TR4 is electrically connected to the first node N1, and a gate electrode of the fourth switching element TR4 is electrically connected to the *i*th emission control line E_{*i*}.

(63) One electrode of the fifth switching element TR5 is electrically connected to the second node N2, and an other electrode of the fifth switching element TR5 is electrically connected to the one end of the light-emitting elements **350**. A gate electrode of the fifth switching element TR5 is electrically connected to the *i*th emission control line E_{*i*}. The fourth switching element TR4 and the

fifth switching element TR5 are turned on simultaneously according to the emission control signal transmitted via the i th emission control line Ei so that the first driving voltage VDD is applied to the light-emitting elements 350, thereby allowing the driving current to flow through the light-emitting elements 350.

(64) One electrode of the sixth switching element TR6 is electrically connected to the third node N3, and an other electrode of the sixth switching element TR6 is supplied with an initializing voltage Vinit. The gate electrode of the sixth switching element TR6 is electrically connected to the $(i-1)$ th scan line SL $(i-1)$. Because the scan signal is sequentially supplied to the scan lines SL, the scan signal may be supplied to the $(i-1)$ th scan line SL $(i-1)$ and then to the i th scan line SL i . The sixth switching element TR6 is turned on in response to the scan signal transmitted via the $(i-1)$ th scan line SL $(i-1)$ to transmit the initializing voltage Vinit to the gate electrode of the first switching element TR1, thereby initializing the voltage of the gate electrode of the first switching element TR1.

(65) One electrode of the seventh switching element TR7 is electrically connected to the one end of the light-emitting elements 350, and an other electrode of the seventh switching element TR7 is supplied with the initializing voltage Vinit. A gate electrode of the seventh switching element TR7 is electrically connected to the $(i-1)$ th scan line SL $(i-1)$. The seventh switching element TR7 may be turned on in response to the scan signal to initialize the one end electrode of the light-emitting elements 350.

(66) The first to seventh switching elements TR1 to TR7 may be thin-film transistors. In each of the first to seventh switching elements TR1 to TR7, one electrode may be a source electrode or a drain electrode, and the other electrode may be the drain electrode or the source electrode.

(67) One end of the storage capacitor Cst may be electrically connected to the first power supply line VDDL to receive the first driving voltage VDD, and the other end may be electrically connected to the third node N3.

(68) Hereinafter, a number of elements disposed on the display device 10 will be described in more detail with reference to FIG. 4.

(69) FIG. 4 is a cross-sectional view taken along the line I-I' of FIG. 1. Although FIG. 4 shows only one pixel PX, it is to be understood that the description can be equally applied to other pixels as well.

(70) Referring to FIGS. 1 and 4, a display device 10 may include a substrate 110, thin-film transistors 120 and 140 disposed on the substrate 110, electrodes 310, 320 and 330 and light-emitting elements 350 disposed above the thin-film transistors 120 and 140. The thin-film transistors may include the first thin-film transistor 120 and the second thin-film transistor 140, which may be a driving transistor and a switching transistor, respectively. Each of the thin-film transistors 120 and 140 may have an active layer, a gate electrode, a source electrode and a drain electrode. The first electrode 310 may be electrically connected to the drain electrode of the first thin-film transistor 120.

(71) More specifically, the substrate 110 may be an insulating substrate. The substrate 110 may be made of an insulating material such as glass, quartz or a polymer resin. Examples of the polymer material may include polyethersulphone (PES), polyacrylate (PA), polyacrylate (PAR), polyetherimide (PEI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polyphenylene sulfide (PPS), polyallylate, polyimide (PI), polycarbonate (PC), cellulose triacetate (CAT), cellulose acetate propionate (CAP), or a combination thereof. The substrate 110 may be either a rigid substrate or a flexible substrate that can be bent, folded, or rolled.

(72) A buffer layer 115 may be disposed on the substrate 110. The buffer layer 115 may prevent impurity ions from diffusing, may prevent permeation of moisture or outside air, and may provide a flat surface. The buffer layer 115 may include silicon nitride, silicon oxide, silicon oxynitride, or the like.

(73) A semiconductor layer is disposed on the buffer layer 115. The semiconductor layer may

include a first active layer **126** of the first thin-film transistor **120**, a second active layer **146** of the second thin-film transistor **140**, and an auxiliary layer **163**. The semiconductor layer may include polycrystalline silicon, monocrystalline silicon, an oxide semiconductor, etc.

(74) A first gate insulating layer **170** is disposed on the semiconductor layer. The first gate insulating layer **170** covers the semiconductor layer. The first gate insulating layer **170** may work as a gate insulating layer of the thin-film transistors. The first gate insulating layer **170** may include silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, tantalum oxide, hafnium oxide, zirconium oxide, titanium oxide, etc. They may be used alone or in combinations.

(75) A first conductive layer is disposed on the first gate insulating layer **170**. The first conductive layer may include a first gate electrode **121**, a second gate electrode **141** and a power supply line **161** that are disposed on the first active layer **126** of the first thin-film transistor **120**, on the second active layer **146** of the second thin-film transistor **140**, and on the auxiliary layer **163**, respectively, with the first gate insulating layer **170** therebetween. The first conductive layer may include at least one metal selected from the group consisting of: molybdenum (Mo), aluminum (Al), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium (Ti), tantalum (Ta), tungsten (W) and copper (Cu). The first conductive layer may be made up of a single layer or multiple layers.

(76) A second gate insulating layer **180** is disposed on the first conductive layer. The second gate insulating layer **180** may be an interlayer dielectric layer. The second gate insulating layer **180** may be made of an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, hafnium oxide, aluminum oxide, titanium oxide, tantalum oxide or zinc oxide.

(77) A second conductive layer is disposed on the second gate insulating layer **180**. The second conductive layer includes a capacitor electrode **128** disposed above the first gate electrode **121** with a second insulating layer therebetween. The capacitor electrode **128** may form a storage capacitor with the first gate electrode **121**.

(78) The second conductive layer may include at least one metal selected from the group consisting of: molybdenum (Mo), aluminum (Al), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium (Ti), tantalum (Ta), tungsten (W) and copper (Cu), like the first conductive layer described above.

(79) An interlayer dielectric layer **190** is disposed on the second conductive layer. The interlayer dielectric layer **190** may be an interlayer dielectric film. The interlayer dielectric layer **190** may also provide a flat surface. The interlayer dielectric layer **190** may include an organic insulating material such as polyacrylate resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, unsaturated polyesters resin, poly phenylene ether resin, poly phenylene sulfide resin, and benzocyclobutene (BCB).

(80) A third conductive layer is disposed on the interlayer dielectric layer **190**. The third conductive layer includes the first drain electrode **123** and the first source electrode **124** of the first thin-film transistor **120**, the second drain electrode **140** and the second source electrode **144** of the second thin-film transistor **140**, and a power supply electrode **162** disposed on the power supply line **161**.

(81) The first source electrode **124** and the first drain electrode **123** may be electrically connected to the first active layer **126** through the first contact holes **129** penetrating the interlayer dielectric layer **190** and the second gate insulating layer **180**, respectively. The second source electrode **144** and the second drain electrode **143** may be electrically connected to the second active layer **146** through the second contact holes **149** penetrating the interlayer dielectric layer **190** and the second gate insulating layer **180**, respectively. The power supply electrode **162** may be electrically connected to the power supply line **161** through a third contact hole **169** penetrating the interlayer dielectric layer **190** and the second gate insulating layer **180**.

(82) The third conductive layer may include at least one metal selected from the group consisting of: aluminum (Al), molybdenum (Mo), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium

(Ti), tantalum (Ta), tungsten (W) and copper (Cu). The third conductive layer may be made up of a single film or multiple films. For example, the third conductive layer may have a stack structure of Ti/Al/Ti, Mo/Al/Mo, Mo/AlGe/Mo, and Ti/Cu.

(83) An insulating substrate layer **300** is disposed over the third conductive layer. The insulating substrate layer **300** may be made of an organic insulating material such as polyacrylate resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, unsaturated polyesters resin, poly phenylene ether resin, poly phenylene sulfide resin, and benzocyclobutene (BCB). The surface of the insulating substrate layer **300** may be flat.

(84) A plurality of pixel-defining layers **410**, **420** and **430** may be disposed on the insulating substrate layer **300**. The pixel-defining layers **410**, **420** and **430** may be disposed in each of the pixels PX and may be spaced apart from one another. The first electrode **310**, the second electrode **320** and the third electrode **330** may be disposed on the first pixel-defining layer, **410**, the second pixel-defining layer **420** and the third pixel-defining layer **430**, respectively. In the examples shown in FIGS. **1** and **4**, three pixel-defining layers **410**, **420** and **430** are disposed in one pixel PX, and the first electrode **310**, the second electrode **320** and the third electrode **330** are arranged.

(85) It is, however, to be understood that the present disclosure is not limited thereto. A greater number of pixel-defining layers **410**, **420** and **430** may be disposed in one pixel PX. For example, when a greater number of third electrodes **330** are disposed between the first electrode **310** and the second electrode **320**, the pixel PX may have a greater number of pixel-defining layers. The pixel-defining layers **410**, **420** and **430** may include at least one first pixel-defining layer **410** on which the first electrode **310** is disposed, at least one second pixel-defining layer **420** on which the second electrode **320** is disposed, and at least one third pixel-defining layer **430** on which the third electrode **330** is disposed. In such a case, the first pixel-defining layer **410** and the second pixel-defining layer **420** may be spaced apart from each other, and a plurality of third pixel-defining layer **430** may be disposed therebetween. For another example, the first pixel-defining layer **410**, the second pixel-defining layer **420** and the third pixel-defining layer **430** may be alternately arranged in one direction. In some exemplary embodiments, two first pixel-defining layers **410** may be disposed apart from each other, one second pixel-defining layer **420** may be disposed between the first pixel-defining layers **410**, and at least one third pixel-defining layer **430** may be disposed between the second pixel-defining layer **420** and each of the first pixel-defining layers **410**.

(86) The plurality of pixel-defining layers **410**, **420** and **430** may be made of substantially the same material via a single process. In such a case, the pixel-defining layer ribs **410**, **420** and **430** may form a single lattice pattern. The pixel-defining layers **410**, **420** and **430** may include polyimide (PI).

(87) Although not shown in the drawings, at least some of the plurality of pixel-defining layers **410**, **420** and **430** may be disposed at the boundaries between the pixels PX so as to distinguish one from another. Such pixel-defining layers may also be disposed in a substantially lattice pattern together with the first pixel-defining layer **410**, the second pixel-defining layer **420**, and the third pixel-defining layer **430** described above. At least some of the pixel-defining layers **410**, **420** and **430** disposed at the boundaries between the pixels PX may be formed to cover the electrode lines of the display device **10**.

(88) Reflective layers **311**, **321** and **331** may be disposed on the pixel-defining layers **410**, **420** and **430**.

(89) The first reflective layer **311** covers the first pixel-defining layer **410** and is electrically connected to the first drain electrode **123** of the first thin-film transistor **120** through a fourth contact hole **319_1** penetrating the insulating substrate layer **300**. The second reflective layer **321** covers the second pixel-defining layer **420** and is electrically connected to the power supply electrode **162** through a fifth contact hole **319_2** penetrating through the insulating substrate layer **300**. The third reflective layer **331** covers the third pixel-defining layer **430**, and both ends thereof are spaced apart from the first reflective layer **311** and the second reflective layer **321**. Unlike the

first reflective layer **311** and the second reflective layer **321**, the third reflective layer **331** is connected to none of the thin-film transistors through a contact hole.

(90) The first reflective layer **311** may be electrically connected to the first drain electrode **123** of the first thin-film transistor **120** through the fourth contact hole **319_1** in the pixel PX. Accordingly, the first thin-film transistor **120** may be disposed in an area overlapping the pixel PX. In the example shown in FIG. **1**, the first reflective layer **311** is electrically connected to the first thin-film transistor **120** through the first electrode contact hole CNTD formed on the first electrode stem **310S**. That is to say, the first electrode contact hole CNTD may be the fourth contact hole **319_1**.

(91) The second reflective layer **321** may also be electrically connected to the power supply electrode **162** through the fifth contact hole **319_2** in the pixel PX. In FIG. **4**, the second reflective layer **321** is connected through the fifth contact hole **319_2** in one pixel PX. FIG. **1** illustrates that the second electrode **320** of each pixel PX is electrically connected to the power supply line **161** through a plurality of second electrode contact holes CNTS on the second electrode stem **320S**. That is to say, the second electrode contact hole CNTS may be the fifth contact hole **319_2**. It is, however, to be understood that the present disclosure is not limited thereto. For example, in FIG. **1**, the second electrode contact hole CNTS may be formed at various positions on the second electrode stem **320S**, and on some implementations, may be located on the second electrode branch **320B**. Further, in some exemplary embodiments, the second reflective layer **321** may be connected to one second electrode contact hole CNTS or the fifth contact hole **319_2** in an area other than one pixel PX.

(92) FIG. **5** is a plan view of a display device according to another exemplary embodiment of the present disclosure. FIG. **6** is a cross-sectional view taken along the lines II-II' and III-III' of FIG. **5**.

(93) There may be a non-emission area where no light-emitting element **350** is disposed, e.g., in an area other than the emission portion where the pixels PX of the display device **10** are disposed, for example, on the outer side of the pixels PX. As described above, the second electrodes **320** of each pixel PX may be electrically connected to each other through the second electrode stem **320S** to receive the same electric signal.

(94) Referring to FIGS. **5** and **6**, in some exemplary embodiments, the second electrode stem **320S** of the second electrode **320** may be electrically connected to the power supply electrode **162** through a second electrode contact hole CNTS in the non-emission area located on the outer side portion of the display device **10**. Unlike the display device **10** shown in FIG. **1**, even though the second electrode stem **320S** is connected to the power supply electrode **162** through one contact hole, because the second electrode stem **320S** is extended to and electrically connected to the adjacent pixel PX, it is possible to apply the same electrical signal to the second electrode branch **320B** of each pixel PX. For the second electrode **320** of the display device **10**, the position of the contact hole for receiving the electric signal from the power supply electrode **162** may vary according to the structure of the display device **10**. However, this is merely illustrative.

(95) Referring back to FIGS. **1** and **4**, the reflective layers **311**, **321** and **331** may include a material having a high reflectivity to reflect light emitted from the light-emitting elements **350**. For example, the reflective layers **311**, **321** and **331** may include, but is not limited to, materials such as silver (Ag) and copper (Cu).

(96) At least a part of the plurality of pixel-defining layers **410**, **420** and **430** may protrude from the insulating substrate layer **300**. The pixel-defining layers **410**, **420** and **430** may protrude upwardly from the plane on which the light-emitting elements **350** are disposed, and at least a part of the protruding portions may have an inclination. The pixel-defining layers **410**, **420** and **430** protruding with an inclination can reflect light incident on the reflective layers **311**, **321** and **331** disposed thereon. Light directed from the light-emitting elements **350** to the reflective layers **311**, **321** and **331** may be reflected off them and transmitted toward the outside of the display device **10**, for example, above the pixel-defining layers **410**, **420** and **430**.

(97) The first electrode layer **312**, the second electrode layer **322** and the third electrode layer **332**

may be disposed on the first reflective layer **311**, the second reflective layer **321**, and the third reflective layer **331**, respectively.

(98) The first electrode layer **312** is disposed directly on the first reflective layer **311**. The first electrode layer **312** may have substantially the same pattern as the first reflective layer **311**. The second electrode layer **322** is disposed directly on the second reflective layer **321**. The second electrode layer **322** may have substantially the same pattern as the second reflective layer **321**. The third electrode layer **332** is disposed directly on the third reflective layer **331** and is spaced apart from the first electrode layer **312** and the second electrode layer **322**. The third electrode layer **332** may have substantially the same pattern as the third reflective layer **331**.

(99) In an exemplary embodiment, the electrode layers **321**, **322** and **332** may cover the reflective layers **311**, **321** and **331** thereunder, respectively. That is to say, the electrode layers **321**, **322** and **332** may be larger than the reflective layers **311**, **321** and **331** to cover the side surfaces of the ends of the electrode layers **321**, **322**, and **332**. It is, however, to be understood that the present disclosure is not limited thereto.

(100) The first electrode layer **312** and the second electrode layer **322** may transmit electrical signals transmitted to the first reflective layer **311** and the second reflective layer **321** connected to the first thin-film transistor **120** and the power supply electrode **162**, respectively, to the contact electrodes to be described below. The electrode layers **312**, **322** and **332** may include a transparent conductive material. For example, the electrode layers **312**, **322** and **332** may include, but is not limited to, a material such as indium tin oxide (ITO), indium zinc oxide (IZO) and indium tin zinc oxide (ITZO). In some exemplary embodiments, each of the reflective layers **311**, **321** and **331** and the electrode layers **312**, **322** and **332** may have a structure in which at least one transparent conductive layer such as ITO, IZO and ITZO and at least one metal layer such as silver and copper are stacked on one another. For example, the reflective layers **311**, **321** and **331** and the electrode layers **312**, **322** and **332** may form a stack structure of ITO/silver (Ag)/ITO.

(101) The first reflective layer **311** and the first electrode layer **312**, which are disposed on the first pixel-defining layer **410**, form the first electrode **310**. The first electrode **310** may protrude to areas extended from both ends of the first pixel-defining layer **410** so that the first electrode **310** may be in contact with the insulating substrate layer **300**. The second reflective layer **321** and the second electrode layer **322**, which are disposed on the second pixel-defining layer **420**, form the second electrode **320**. The second electrode **320** may protrude to areas extended from both ends of the second pixel-defining layer **420** so that the second electrode **320** may be in contact with the insulating substrate layer **300**. The third reflective layer **331** and the third electrode layer **332**, which are disposed on the third pixel-defining layer **430**, form the third electrode **330**. The third electrode **330** may protrude to areas extended from both ends of the third pixel-defining layer **430** so that the third electrode **330** may be in contact with the insulating substrate layer **300**.

(102) The first electrode **310**, the second electrode **320** and the third electrode **330** are arranged to cover the entire area of the first pixel-defining layer **410**, the second pixel-defining layer **420** and the third pixel-defining layer **430**. As described above, the first electrode **310** may be spaced apart from and face the third electrode **330**, and the third electrode **330** may be spaced apart from and face the second electrode **320**. A first insulating layer **381** may be disposed between the electrodes spaced apart from one another, which will be described in detail below, and the light-emitting elements **350** may be disposed on the first insulating layer **381**.

(103) Because the first reflective layer **311** may receive a driving voltage from the first thin-film transistor **120** and the second reflective layer **321** may receive a supply voltage from the power supply line **161**, the first electrode **310** and the second electrode **320** receive the driving voltage and the supply voltage, respectively. As will be described below, the first electrode **310** may be electrically connected to the first thin-film transistor **120**, and the second electrode **320** may be electrically connected to the power supply line **161**. Accordingly, a first contact electrode **361** and a second contact electrode **362** disposed on the first electrode **310** and the second electrode **320**,

respectively, can receive the driving voltage and the supply voltage. The third electrode **330** disposed between the first electrode **310** and the second electrode **320** may transmit the driving voltage and the supply voltage to the light-emitting elements **350** through a third contact electrode **363** disposed thereon, such that a current flows in the light-emitting elements **350** to emit light.

(104) The first insulating layer **381** is disposed on a part of the first electrode **310**, the second electrode **320** and the third electrode **330**. The first insulating layer **381** may be disposed in the space between the first electrode **310** and the third electrode **330** and between the third electrode **330** and the second electrode **320**. The first insulating layer **381** may have an island shape or a linear shape formed along the space between the first electrode branch **310B**, the second electrode branch **320B**, and the third electrode **330** when viewed from the top.

(105) The light-emitting elements **350** are disposed on the first insulating layer. The first insulating layer **381** may be disposed between the light-emitting elements **350** and the insulating substrate layer **300**. The lower surface of the first insulating layer **381** may be in contact with the insulating substrate layer **300**, and the light-emitting elements **350** may be disposed on the upper surface of the first insulating layer **381**. The first insulating layer **381** is in contact with the electrodes **310**, **320**, and **330** on both sides and can electrically insulate them from each other.

(106) The first insulating layer **381** may overlap with a portion of each of the electrodes **310**, **320** and **330**, for example, a region protruding in a direction in which the first electrode **310** and the third electrode **330** face each other, and a region protruding in a direction in which the third electrode **330** and the second electrode **320** face each other. In addition, the first insulating layer **381** may be disposed on the inclined side surfaces of the pixel-defining layers **410**, **420** and **430** among the regions on the electrodes **310**, **320** and **330**. Accordingly, the lower surface of the first insulating layer **381** may be in contact with the insulating substrate layer **300** and the electrodes **310**, **320** and **330**. At least one side portion of the first insulating layer **381** may be in contact with each of the electrodes **310**, **320** and **330** as well. A part of the upper surface of the electrodes **310**, **320** and **330** where the first insulating layer **381** is not disposed may be exposed. In addition, the first insulating layer **381** may be disposed such that both sides of the light-emitting elements **350** are exposed. Accordingly, the contact electrodes **361**, **362** and **363** described below may be in contact with the exposed upper surfaces of the electrodes **310**, **320** and **330** and both sides of the light-emitting elements **350**.

(107) For example, the first insulating layer **381** may cover the upper surface of the regions where the first electrode **310** faces the third electrode **330**, and the third electrode **330** faces the second electrode **320**. The first insulating layer **381** protects the overlapped regions with the electrodes **310**, **320** and **330** and electrically insulates them from each other. In addition, the first insulating layer **381** prevents a first conductivity-type semiconductor layer **351** and the second conductivity-type semiconductor layer **352** of the light-emitting element **350** from being in contact with other substrates, so that damage to the light-emitting element **350** can be avoided.

(108) At least one light-emitting element **350** may be disposed between the first electrode **310** and the third electrode **330** and between the third electrode **330** and the second electrode **320**.

Accordingly, the plurality of light-emitting elements **350** may be connected in series. FIG. 1 illustrates an example where only the light-emitting elements **350** that emit light of the same color are disposed in each pixel PX. It is to be understood that the light-emitting elements **350** for emitting light of different colors may be disposed in one pixel PX, as mentioned earlier.

(109) The distance between the first electrode **330** and the third electrode **330**, and the distance between the third electrode **330** and the second electrode **320** may be less than or equal to the length of the light-emitting elements **350**, as described above. Accordingly, an electrical contact between the light-emitting elements **350** and each of the first electrode **310**, the second electrode **320**, and the third electrode **330** can be made easily and reliably.

(110) The light-emitting elements **350** may be light-emitting diodes. The light-emitting elements **350** may be a nanostructure generally having its size in a nanometer scale. The light-emitting

elements **350** may be inorganic light-emitting diodes made of an inorganic material. When the light-emitting element **350** is an inorganic light-emitting diode, by disposing a light-emitting material having an inorganic crystal structure between two opposing electrodes and applying an electric field in a direction across the light-emitting material, the inorganic light-emitting diodes can be aligned between the two electrodes having the polarities.

(111) Referring to the enlarged circle of FIG. **4**, in some exemplary embodiments, the light-emitting element **350** may include a first conductivity-type semiconductor layer **351**, an active material layer **353**, a second conductivity-type semiconductor layer **352** and an electrode material layer **357**, which will be described in detail below. The light-emitting element **350** may be formed by stacking the first conductivity-type semiconductor layer **351**, the active material layer **353**, the second conductivity-type semiconductor layer **352** and the electrode material layer **357** in parallel with the insulating substrate layer **300**. In other words, the light-emitting element **350**, in which the plurality of layers are stacked, may be disposed in the horizontal direction in parallel with the insulating substrate layer **300**. It is, however, to be understood that the present disclosure is not limited thereto. The layers of the light-emitting element **350** may be stacked on one another in the opposite order.

(112) In an exemplary embodiment, the first conductivity-type semiconductor layer **351** of the light-emitting element **350** may be electrically connected to a side of the first electrode **310** or the third electrode **330** that faces the second electrode **320**, and the second conductivity-type semiconductor layer **352** or the electrode material layer **357** of the light-emitting element **350** may be electrically connected to the other side of the third electrode **330** or the second electrode **320**. The configuration of the light-emitting element **350** will be described in more detail later.

(113) The second insulating layer **382** may be disposed on the light-emitting element **350** to protect the light-emitting element **350** and to fix it between the electrodes **310**, **320** and **330**. Although not shown in FIG. **4**, the second insulating layer **382** may be disposed on the outer surface of the light-emitting elements **350** to fix them. The second insulating layer **382** may be disposed on a part of the outer surface of the light-emitting element **350** so that both side surfaces of the light-emitting element **350** are exposed. Specifically, the length of the second insulating layer **382** is smaller than that of the light-emitting element **350**, so that the second insulating layer **382** may be recessed inwardly from both sides of the light-emitting element **350**. Accordingly, the side surfaces of the first insulating layer **381**, the light-emitting element **350** and the second insulating layer **382** may have a stair-like shape. In such a case, the second insulating layer **382** is disposed like the first insulating layer **381**, so that the contact electrodes **361**, **362** and **363** can be brought into contact with the side surfaces of the light-emitting element **350** easily and reliably. It is, however, to be understood that the present disclosure is not limited thereto. The length of the second insulating layer **382** may be equal to the length of the light-emitting element **350** and the both sides may be aligned with each other.

(114) On the second insulating layer **382**, the first contact electrode **361** disposed on the first electrode **310** and overlapping with at least a part of the second insulating layer **382**, the second contact electrode **362** disposed on the second electrode **320** and overlapping with at least a part of the second insulating layer **382**, and the third contact electrode **363** disposed on the third electrode **330** and spaced apart from the first contact electrode **361** and the second contact electrode **362** may be disposed.

(115) The first contact electrode **361** and the second contact electrode **362** may be disposed on the upper surfaces of the first electrode **310** and the second electrode **320**, respectively. Specifically, the first contact electrode **361** and the second contact electrode **362** may be disposed on the upper surfaces of the first electrode **310** and the second electrode **320** to be in contact with the first electrode layer **312** and the second electrode layer **322**, respectively. The first contact electrode **361** and the second contact electrode **362** may be in contact with the first conductivity-type semiconductor layer **351** and the second conductivity-type semiconductor layer **352** of the light-

emitting element **350**, respectively. Accordingly, the first contact electrode **361** and the second contact electrode **362** may transmit electric signals applied to the first electrode layer **312** and the second electrode layer **322** to the light-emitting element **350**.

(116) The first contact electrode **361** may be disposed on and partially cover the first electrode **310**. The lower surface of the first contact electrode **361** may be in contact with a part of each of the light-emitting element **350**, the first insulating layer **381** and the second insulating layer **382**. One end of the first contact electrode **361** closer to the third electrode **330** is disposed on the second insulating layer **382**. The second contact electrode **362** may be disposed on and partially cover the second electrode **320**. The lower surface of the second contact electrode **362** may be in contact with a part of each of the light-emitting element **350**, the first insulating layer **381** and the second insulating layer **382**. One end of the second contact electrode **362** closer to the third electrode **330** is disposed on the second insulating layer **382**.

(117) As shown in FIG. 1, neither first contact electrode **361** nor the second contact electrode **362** is disposed on the first electrode contact hole CNTD formed in the first electrode stem **310S** and the second electrode contact hole CNTS formed in the second electrode stem **320S**. That is to say, the first contact electrode **361** and the second contact electrode **362** may not overlap with the first electrode contact hole CNTD and the second electrode contact hole CNTS, respectively. It is, however, to be understood that the present disclosure is not limited thereto. In some implementations, a part of each of the first contact electrode **361** and the second contact electrode **362** may be disposed on the first electrode **310** and the second electrode **320** such that they overlap with the first electrode contact hole CNTD and the second electrode contact hole CNTS, respectively.

(118) The third contact electrode **363** may be disposed on and partially cover the third electrode **330**. The lower surface of the third contact electrode **363** may be in contact with a part of each of the light-emitting element **350**, the first insulating layer **381** and the third insulating layer **383**. Both ends of the third contact electrode **363** closer to the first electrode **310** and the second electrode **320**, respectively, are disposed on the third insulating layer **383**.

(119) The first contact electrode **361** and the second contact electrode **362** may be apart from the third contact electrode **363** on the second insulating layer **382** or the third insulating layer **383**. Specifically, the first contact electrode **361**, the second contact electrode **362** and the third contact electrode **363** are in contact with the second insulating layer **382** or the third insulating layer **383** together with the light-emitting element **350**, but are spaced apart from one another in the stacking direction and not connected to one another on the second insulating layer **382**. Accordingly, the first contact electrode **361** and the second contact electrode **362** may receive different power from the first thin-film transistor **120** and the power supply line **161**. For example, the first contact electrode **361** may receive a driving voltage applied to the first electrode **310** from the first thin film transistor **120**, and the second contact electrode **362** may receive a common supply voltage applied to the second electrode **320** from the power supply line **161**. It is, however, to be understood that the present disclosure is not limited thereto.

(120) The contact electrodes **361**, **362** **363** may include a conductive material. For example, the contact electrodes may include ITO, IZO, ITZO, aluminum (Al), etc. It is, however, to be understood that the present disclosure is not limited thereto.

(121) In addition, the contact electrodes **361**, **362** and **363** may include the same material as the electrode layers **312**, **322** and **332**. The contact electrodes **361**, **362** and **363** may be arranged in substantially the same pattern on the electrode layers **312**, **322** and **332** so as to be in contact with the electrode layers **312**, **322** and **332**. For example, the first contact electrode **361** and the second contact electrode **362** which are in contact with the first electrode layer **312** and the second electrode layer **322**, respectively, receive electrical signals applied to the first electrode layer **312** and the second electrode layer **322** to transmit them to the light-emitting element **350**.

(122) The third insulating layer **383** may be disposed on the first contact electrode **361** and the

second contact electrode **362** so that the first contact electrode **361** and the second contact electrode **362** are electrically insulated from the third contact electrode **363**. The third insulating layer **383** is disposed to cover the first contact electrode **361** and the second contact electrode **362** but does not overlap with a part of the light-emitting element **350** so that the light-emitting element **350** can be in contact with the third contact electrode **363**. The third insulating layer **383** may be partially in contact with the first contact electrode **361**, the second contact electrode **362** and the second insulating layer **382** on the upper surface of the second insulating layer **382**. The third insulating layer **383** may be disposed on the upper surface of the second insulating layer **382** so that it covers one end of the first contact electrode **361** and the second contact electrode **362**. Accordingly, the third insulating layer **383** may protect the first contact electrode **361** and the second contact electrode **362** and electrically insulate them from the third contact electrode **363**.

(123) One end of the third insulating layer **383** closer to the third electrode **330** may be aligned with one side of the second insulating layer **382**.

(124) In some exemplary embodiments, the third insulating layer **383** may be eliminated from the display device **10**. Then, the first contact electrode **361**, the second contact electrode **362**, and the third contact electrode **363** may be disposed on substantially the same plane, and the first contact electrode **361**, the second contact electrode **362** and the third contact electrode **363** may be electrically insulated from one another by a passivation layer **385**, which will be described later. More detailed description thereon will be given with reference to other exemplary embodiments.

(125) The passivation layer **385** may be formed over the third insulating layer **383** and the third contact electrode **363** to protect elements disposed on the insulating substrate layer **300** from the external environment. If the first contact electrode **361**, the second contact electrode **362** and the third contact electrode **363** are exposed, the electrodes may be damaged and thus there may arise a problem of disconnection. To prevent this, the passivation layer **385** is disposed to cover them. That is to say, the passivation layer **385** may be disposed to cover the first electrode **310**, the second electrode **320**, the third electrode **330**, the light-emitting element **350**, etc. If the third insulating layer **383** is eliminated as described above, the passivation layer **385** may be formed on the first contact electrode **361**, the second contact electrode **362** and the third contact electrode **363**. In such a case, the passivation layer **385** may electrically insulate the first contact electrode **361**, the second contact electrode **362** and the third contact electrode **363** from one another.

(126) Each of the first insulating layer **381**, the second insulating layer **382**, the third insulating layer **383** and the passivation layer **385** may include an inorganic insulating material. For example, the first insulating layer **381**, the second insulating layer **382**, the third insulating layer **383**, and the passivation layer **385** may include a material such as silicon oxide (SiO_x), silicon nitride (SiN_x), silicon oxynitride (SiO_xN_y), aluminum oxide (Al₂O₃) and aluminum nitride (AlN). The first insulating layer **381**, the second insulating layer **382**, the third insulating layer **383** and the passivation layer **385** may be formed of either the same material or different materials. Besides, each of the first insulating layer **381**, the second insulating layer **382**, the third insulating layer **383** and the passivation layer **385** may be made of any of a variety of materials that have insulation properties.

(127) The light-emitting element **350** may be formed on a substrate by an epitaxial technique. It may be grown by forming a seed crystal layer for forming a semiconductor layer on a substrate and then depositing a desired semiconductor material. Hereinafter, the structure of the light-emitting element **350** according to various exemplary embodiments will be described in detail with reference to FIGS. **7** to **9**.

(128) FIG. **7** is a view showing the light-emitting element according to an exemplary embodiment of the present disclosure.

(129) Referring to FIG. **7**, the light-emitting element **350** may include a plurality of conductive material layers **351** and **352**, an active material layer **353** disposed between the plurality of conductive material layers **351** and **352**, an electrode material layer **357**, and an insulating material

layer **358**. An electric signal applied from the first electrode **310** and the second electrode **320** may be transmitted to the active material layer **353** via the plurality of conductive material layers **351** and **352** to emit light.

(130) Specifically, the light-emitting element **350** may include the first conductivity-type semiconductor layer **351**, the second conductivity-type semiconductor layer **352**, the active material layer **353** disposed between the first conductivity-type semiconductor layer **351** and the second conductivity-type semiconductor layer **352**, the electrode material layer **357** disposed on the second conductivity-type semiconductor layer **352**, and the insulating material layer **358**. In the light-emitting element **350** shown in FIG. 7, the first conductivity-type semiconductor layer **351**, the active material layer **353**, the second conductivity-type semiconductor layer **352** and the electrode material layer **357** are stacked on one another in the longitudinal direction in this order. It is, however, to be understood that the present disclosure is not limited thereto. The electrode material layer **357** may be eliminated. In some exemplary embodiments, it may be disposed on at least one of the first and second conductivity-type semiconductor layers **351** and **352**. Hereinafter, the light-emitting element **350** of FIG. 7 will be described as an example.

(131) The first conductivity-type semiconductor layer **351** may be an n-type semiconductor layer. For example, when the light-emitting element **350** emits light of a blue wavelength band, the first conductivity-type semiconductor layer **351** may be a semiconductor material having Formula below: $\text{In}_x\text{Al}_y\text{Ga}_{1-x-y}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). For example, it may be at least one of n-type doped InAlGaN , GaN , AlGaN , InGaN , AlN and InN . The first conductivity-type semiconductor layer **351** may be doped with a first conductivity-type dopant. For example, the first conductivity-type dopant may be Si, Ge, Sn, etc. The length of the first conductivity-type semiconductor layer **351** may range, but is not limited to, from 1.5 μm to 5 μm .

(132) The second conductivity-type semiconductor layer **352** may be a p-type semiconductor layer. For example, when the light-emitting element **350** emits light of a blue wavelength band, the first conductivity-type semiconductor layer **352** may be a semiconductor material having Formula below: $\text{In}_x\text{Al}_y\text{Ga}_{1-x-y}\text{N}$ ($0 \leq x \leq 2$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). For example, it may be at least one of p-type doped InAlGaN , GaN , AlGaN , InGaN , AlN and InN . The second conductivity-type semiconductor layer **352** may be doped with a second conductivity-type dopant. For example, the second conductivity-type dopant may be Mg, Zn, Ca, Se, Ba, etc. The length of the second conductivity-type semiconductor layer **352** may range, but is not limited to, from 0.08 μm to 0.25 μm .

(133) The active material layer **353** may be disposed between the first conductivity-type semiconductor layer **351** and the second conductivity-type semiconductor layer **352** and may include a material having a single or multiple quantum well structure. When the active material layer **353** includes a material having the multiple quantum well structure, quantum layers and well layers may be alternately stacked in the structure. The active material layer **353** may emit light as electron-hole pairs are combined therein in response to an electrical signal applied through the first conductivity-type semiconductor layer **351** and the second conductivity-type semiconductor layer **352**. For example, when the active material layer **353** emits light in the blue wavelength range, the active material layer **353** may include a material such as AlGaN and AlInGaN . In particular, when the active material layer **353** has the multiple quantum well structure in which the quantum layers and the well layers are stacked on one another alternately, the quantum layers may include AlGaN or AlInGaN , and the well layers may include materials such as GaN or AlGaN . It is, however, to be understood that the present disclosure is not limited thereto. The active material layer **353** may have a structure in which a semiconductor material having a large band gap energy and a semiconductor material having a small band gap energy are alternately stacked on one another, and may include other Group III to Group V semiconductor materials depending on the wavelength range of the emitted light. Accordingly, the light emitted from the active material layer **353** is not limited to the light of the blue wavelength band. The active material layer **353** may emit light of red or green wavelength band in some implementations. The length of the active material layer **353**

may be, but is not limited to, in the range of 0.05 μm to 0.25 μm .

(134) The light emitted from the active material layer **353** may exit through not only the outer surfaces of the light-emitting element **350** in the longitudinal direction but also through both side surfaces. That is to say, the direction in which the light emitted from the active material layer **353** propagates is not limited to one direction.

(135) The electrode material layer **357** may be an ohmic contact electrode. It is, however, to be understood that the present disclosure is not limited thereto. The electrode material layer **357** may be a Schottky contact electrode. The electrode material layer **357** may include a metal having conductivity. For example, the electrode material layer **357** may include at least one of aluminum (Al), titanium (Ti), indium (In), gold (Au) and silver (Ag). The electrode material layer **357** may include the same material and may include different materials. It is, however, to be understood that the present disclosure is not limited thereto.

(136) The insulating material layer **358** may be formed outside the light-emitting element **350** to protect it. For example, the insulating material layer **358** may be formed so as to surround the side surface of the light-emitting element **350**, and not at both ends of the light-emitting element **350** in the longitudinal direction, for example, where the first conductivity-type semiconductor layer **351** and the second conductivity semiconductor layer **352** are disposed. It is, however, to be understood that the present disclosure is not limited thereto. The insulating material layer **358** may include materials having an insulating property such as silicon oxide (SiOx), silicon nitride (SiNx), silicon oxynitride (SiOxNy), aluminum nitride (AlN) and aluminum oxide (Al_2O_3). Accordingly, it is possible to prevent an electrical short-circuit that may be created when the active material layer **353** comes in contact with the first electrode **310** or the second electrode **320**. In addition, because the insulating material layer **358** may include the active material layer **353** to protect the outer surface of the light-emitting element **350**, it is possible to prevent a decrease in luminous efficiency.

(137) The insulating material layer **358** may be extended in the longitudinal direction to cover from the first conductivity-type semiconductor layer **351** to the electrode material layer **357**. It is, however, to be understood that the present disclosure is not limited thereto. The insulating material layer **358** may cover only the first conductivity-type semiconductor layer **351**, the active material layer **353** and the second conductivity-type semiconductor layer **352** or only a part of the electrode material layer **357**, such that the other part of the electrode material layer **357** may be exposed.

(138) In some exemplary embodiments, the insulating material layer **358** may be subjected to a surface treatment so that it is dispersed in the solution without being agglomerated with another insulating material layer. During the process of aligning the light-emitting elements **350** to be described below, the light-emitting elements **350** remain dispersed in the solution to be independently aligned between the first electrode **310**, the second electrode **320**, and the third electrode **330**. For example, the surface of the insulating material layer **358** may be treated to be hydrophobic or hydrophilic, so that it can remain dispersed in the solution.

(139) The thickness of the insulating material layer **358** may be, but is not limited to, in the range of 0.5 μm to 1.5 μm .

(140) The light-emitting element **350** may be cylindrical. It is to be understood that the shape of the light-emitting element **350** is not limited thereto but may have various shapes such as a cube, a rectangular cuboid and a hexagonal column. The length of the light-emitting element **350** may range from 1 μm to 10 μm or from 2 μm to 5 μm , and preferably approximately 4 μm . The diameter of the light-emitting element **350** may range from 410 nm to 700 nm, and preferably approximately 500 nm.

(141) Hereinafter, a method of fabricating the display device **10** according to an exemplary embodiment of the present disclosure will be described with reference to FIGS. **8** to **10**.

(142) FIGS. **8** to **10** are plan views schematically showing a method of fabricating a display device according to an exemplary embodiment of the present disclosure.

(143) Referring to FIG. **8**, a plurality of electrodes **310**, **320** and **330** are formed on an insulating

substrate layer **300**. The plurality of electrodes **310**, **320** and **330** may be formed by patterning a metal or an organic material via a typical mask process.

(144) As described above with reference to FIG. **1**, the first electrode stem **310S** and the second electrode stem **320S** are extended in the first direction **D1** and are spaced apart from and located opposite to each other. In FIG. **1**, the first electrode stem **310S** is electrically separated and spaced apart from one another from pixel to pixel **PX**. In contrast, in FIG. **8**, one end of the first electrode stem **310S** may be extended across the plurality of adjacent pixels **PX**. One end of each of the first electrode stem **310S** and the second electrode stem **320S** may be connected to a signal application pad **PAD** so that AC power may be applied to align the light-emitting element **350**.

(145) The first electrode branch **310B** and the second electrode branch **320B** branch off from the first electrode stem **310S** and the second electrode stem **320S**, respectively, and are extended in the second direction **D2**. As described above, the first electrode branch **310B** and the second electrode branch **320B** are extended in opposite directions to each other and are spaced apart from the second electrode stem **320S** and the first electrode stem **310S**, respectively.

(146) The third electrode **330** is spaced apart from the first electrode branch **310B** and the second electrode branch **320B** and is extended in the second direction **D2**. As shown in FIG. **1**, the third electrode **330** is substantially parallel to the first electrode branch **310B** and the second electrode branch **320B**. It is to be noted that because the third electrode **330** is electrically connected neither to the first electrode stem **310S** nor to the second electrode stem **320S**, the AC power applied from the signal application pad **PAD** may not be directly transmitted to it.

(147) That is to say, the first electrode **310** and the second electrode **320** of FIG. **8** are substantially identical to those of FIG. **1** except that the first electrode stem **310S** of each pixel **PX** is extended to be connected to the first electrode stem **310S** of an adjacent pixel. The detailed description thereon will not be given.

(148) Subsequently, referring to FIG. **9**, the light-emitting elements **350** are arranged between the electrodes **310**, **320** and **330**.

(149) Although not shown in the drawings, a solution including the light-emitting elements **350** is applied between the first electrode branch **310B** and the second electrode branch **320B**. Then, AC power is applied from the signal application pad **PAD** connected to one end of each of the first electrode stem **310S** and the second electrode stem **320S** to align the light-emitting elements **350**.

(150) The AC power applied from the signal application pad **PAD** may form a capacitance induced by an electric field between the first electrode branch **310B** and the second electrode branch **320B**. The light-emitting elements **350** in the applied solution may receive a dielectrophoresis force (DEP force) by the capacitance due to the electric field. The light-emitting elements **350** may be aligned between the first electrode branch **310B** and the third electrode **330** and between the third electrode **330** and the second electrode branch **320B** by the DEP force. As the light-emitting elements **350** receive the DEP force, one end of the light-emitting elements **350** is connected to the first electrode branch **310B** or the second electrode branch **320B**, and the other end of the light-emitting elements **350** is connected to the third electrode **330** on both sides.

(151) The dielectrophoresis force applied to the light-emitting elements **350** by the capacitance may cause the light-emitting elements **350** to have a certain directionality between the electrodes **310**, **320** and **330**. For example, the first conductivity-type semiconductor layer **351** of the light-emitting elements **350** may be connected to the first electrode branch **310B** or one side of the third electrode **330** close to the second electrode branch **320B**, and the second conductivity-type semiconductor layer **352** may be connected to the second electrode branch **320B** or the other side of the third electrode **330**. It is, however, to be understood that the present disclosure is not limited thereto. The light-emitting elements **350** may be aligned in any direction between the electrodes **310**, **320** and **330**.

(152) The arrangement of the light-emitting elements **350** have been described above with reference to FIG. **1**.

(153) Subsequently, referring to FIG. 10, after the light-emitting elements 350 are aligned between the electrodes 310, 320 and 330, contact electrodes 360 are formed on the electrodes 310, 320 and 330 and are brought into contact with the light-emitting elements 350. Although not shown in FIG. 10, a plurality of insulating layers may be disposed on the electrodes 310, 320 and 330, as shown in FIG. 4. As described above, the contact electrodes 360 may be in contact with both sides of each of the light-emitting elements 350 and the upper surface of each of the electrodes 310, 320 and 330 on the pixel-defining layers 410, 420 and 430, respectively.

(154) Lastly, as shown in FIG. 10, the first electrode stem 310S is cut at the cutting portions CB so that it is electrically separated, such that the display device 10 of FIG. 1 can be fabricated. The method of electrically separating the first electrode stem 310S is not particularly limited. For example, the portions of the first electrode 310 located at the cutting portions CB may be cut using a laser. As a result, the first electrode stem 310S may be electrically separated and be spaced apart from one another between the neighboring pixels PX.

(155) In addition, the first electrode 310 and the second electrode 312 may be electrically connected to the first thin-film transistor 120 and the power supply electrode 162 through the first electrode contact hole CNTD on the first electrode stem 310S and the second electrode contact hole CNTS on the second electrode stem 320S. Accordingly, a current caused by the driving voltage and the supply voltage may flow between the first electrode 310 and the second electrode 320, such that the light-emitting elements 350 arranged between the first electrode 310 and the third electrode 330 and between the third electrode 330 and the second electrode 320 may be connected in series.

(156) As described above, in the display device 10 according to the exemplary embodiment of the present disclosure, the third electrode 330 is disposed between the first electrode branch 310B and the second electrode branch 320B, such that the light-emitting elements 350 may be aligned between the electrodes to be connected in series. Accordingly, when the display device 10 is driven, the voltage-dividing efficiency can be increased, the capacity of the thin-film transistors can be designed easily, and the loss due to the line resistance can be improved.

(157) Hereinafter, a display device 10 according to another exemplary embodiment of the present disclosure will be described. FIGS. 11 to 13 are plan views of a display device according to another exemplary embodiment of the present disclosure.

(158) Referring to FIGS. 11 and 12, the display device 10 may include a plurality of third electrodes 330 as described above. A display device 10_1 of FIG. 11 may include two third electrodes 330_1 between a first electrode branch 310B_1 and a second electrode branch 320B. Accordingly, the light-emitting elements 350 may be arranged between the first electrode 310_1 and the third electrode 330_1, between the third electrode 330_1 and the third electrode 330_1, and between the third electrode 330_1 and the second electrode 320_1. The number of the light-emitting elements 350 connected in series may be three. A display device 10_2 of FIG. 12 includes six third electrodes 330_2, and the number of the light-emitting elements 350 connected in series may be seven.

(159) Compared to the display device shown in FIG. 1, the number of the light-emitting elements 350 connected in series may vary depending on the number of the third electrodes 330. When an odd number of the third electrode 330 are included in one pixel PX of the display device 10, there may be an even number of the light-emitting elements 350 connected in series in the pixel PX. On the other hand, when an even number of the third electrodes 330 are included, there may be an odd number of the light-emitting elements 350 connected in series.

(160) In other words, when a pixel of the display device 10 includes n third electrodes 330, the number of the light-emitting elements 350 connected in series may be n+1. The number of the light-emitting elements 350 connected in series increases with the number of the third electrodes 330, and thus the voltage-dividing efficiency can be increased and the power loss due to the line resistance can be further improved.

(161) FIG. 13 is a plan view of a display device according to yet another exemplary embodiment of

the present disclosure.

(162) Referring to FIG. 13, a display device **10_3** may include one second electrode **320_3** and two first electrodes **310_3** spaced apart from each other, and the second electrode **320_3** may be disposed between the first electrodes **310_3**. A plurality of third electrodes **330_3** may be disposed between the first electrodes **310_3** and the second electrode **320_3**. Although FIG. 13 shows an example where two third electrodes **330_3** are disposed between the first electrodes **310_3** and the second electrode **320_3**, the present disclosure is not limited thereto.

(163) Specifically, the display device **10_3** may include a plurality of first electrode branches **310B_3** branching off from the first electrode stem **310S_3**. Although two first electrode branches **310B_3** are spaced apart from each other in one pixel PX in the example shown in FIG. 13, the present disclosure is not limited thereto. A greater number of first electrode branches **310B_3** may branch off from it.

(164) One second electrode branch **320_B** may branch off from the second electrode stem **310S_3** in one pixel PX. The second electrode branch **320B_3** may be disposed between the two first electrode branches **310B_3** and may be equally spaced apart from them. In other words, the second electrode branch **320B_3** may be disposed at the center between the first electrode branches **310B_3** spaced apart from each other. Accordingly, one pixel PX of the display device **10_3** may have a laterally symmetrical structure with respect to the second electrode branch **320B_3**. It is, however, to be understood that the present disclosure is not limited thereto. The second electrode branch **320B_3** may be disposed closer to one of the first electrode branches **310B_3**, such that the structure of the pixel may not be symmetrical.

(165) The third electrodes **330_3** may be disposed between the second electrode branch **320B_3** and the two first electrode branches **310B_3**. FIG. 13 shows that two third electrode branches **330B_3** are disposed on either side of the second electrode branch **320B_3** such that four third electrodes **330_3** are disposed in one pixel PX. It is, however, to be understood that the present disclosure is not limited thereto. It is to be understood that a greater number of third electrodes **330_3** may be disposed.

(166) The light-emitting elements **350** may be disposed between the first electrode branches **310B_3** and the third electrodes **330_3**, between the third electrodes **330_3**, and between the third electrodes **330_3** and the second electrode branch **320B_3**. For convenience of illustration, in FIG. 13, the light-emitting elements disposed between the first electrode branch **310B_3** and the third electrodes **330_3** disposed on the left side of the second electrode branch **320B_3** and between the third electrodes **330_3** and the second electrode branch **320B_3** are referred to as a first light-emitting element alignment. The light-emitting elements disposed between the first electrode branch **310B_3** and the third electrodes **330_3** disposed on the right side of the second electrode branch **320B_3** and between the third electrodes **330_3** and the second electrode branch **320B_3** are referred to as a second light-emitting element alignment.

(167) Because one or more third electrodes **330_3** are disposed on the left side of the second electrode branch **320B_3**, the first light-emitting element alignment **350A_1** may include one or more light-emitting element alignments. Because one or more third electrodes **330_3** are disposed on the right side of the second electrode branch **320B_3**, the second light-emitting element alignment **350A_2** may include one or more light-emitting element alignments. In each of the first light-emitting element alignment **350A_1** and the second light-emitting element alignment **350A_2**, one or more light-emitting elements may be connected in series. A current can flow from the first electrode branch **310B_3** to the second electrode branch **320B_3** via the third electrodes **330_3** in the light-emitting element alignment connected in series.

(168) On the other hand, the first light-emitting element alignment **350A_1** and the second light-emitting element alignment **350A_2** disposed on the left and right sides of the second electrode branch **320B_3**, respectively, may be connected to each other in parallel. Because the current flows from different first electrode branches **310B_3** to the same second electrode branch **320B_3** in each

of the first light-emitting element alignment **350A_1** and the second light-emitting element alignment **350A_2**, they may be connected in parallel. In other words, according to this exemplary embodiment of the present disclosure, the display device **10_3** may have the light-emitting element alignment of a series-parallel structure.

(169) As described above, the third electrodes **330** may be substantially identical to the first electrode branches **310B** or the second electrode branch **320B**. In some exemplary embodiments, the third electrodes **330** may be formed by electrically separating the first electrode branches **310B** and the second electrode branches **320B** from the first electrode stem **310S** and the second electrode stem **320S**, respectively. More detailed description thereon will be given with reference to FIGS. **14** and **19**.

(170) FIG. **14** is a plan view of a display device according to yet another exemplary embodiment of the present disclosure.

(171) Referring to FIG. **14**, a display device **10_4** may include a first electrode stem **310S_4**, a second electrode stem **320S_4**, a first electrode branch **310B_4**, a second electrode branch **320B_4**, and a third electrode **330_4**. The arrangements and structures of the first electrode stem **310S_4**, the second electrode stem **320S_4**, the first electrode branch **310B_4** and the second electrode branch **320B_4** are identical to those described with reference to FIG. **1**. The following description will focus on differences.

(172) The third electrode **330_4** may include a third electrode branch **330B_4** and a third electrode fragment **330G_4**.

(173) The third electrode fragment **330G_4** may include the (3-1)th electrode fragment **330Ga_4** branching off from the first electrode stem **310S_4** and terminated while being spaced apart from the third electrode branch **330B_4**, and the (3-2)th electrode fragment **330Gb_4** branching off from the second electrode stem **320S_4** and terminated while being spaced apart from the third electrode branch **330B_4**.

(174) The third electrode branch **330B_4** may include the (3-1)th electrode branch **330Ba_4** spaced apart from the (3-1)th electrode fragment **330Ga_4** and extended in the second direction **D2** to be aligned with the (3-1)th electrode fragment **330Ga_4**, and the (3-2)th electrode branch **330Bb_4** spaced apart from the (3-2)th electrode fragment **330Gb_4** and extended in the second direction **D2** to be aligned with the (3-2)th electrode fragment **330Gb_4**.

(175) In summary, the (3-1)th electrode branch portion **330Ba_4** may be extended in the second direction **D2** and may be terminated while being spaced apart from the (3-1)th electrode fragment **330Ga_4** and the second electrode stem **320S_4**. The (3-1)th electrode branch **330Ba_4** and the (3-1)th electrode fragment **330Ga_4** may be extended and aligned with each other. The (3-2)th electrode branch **330Bb_4** may be extended in the second direction **D2** and may be terminated while being spaced apart from the (3-2)th electrode fragment **330Gb_4** and the first electrode stem **310S_4**. The (3-2)th electrode branch **330Bb_4** and the (3-2)th electrode fragment **330Gb_4** may be extended and aligned with each other.

(176) The third electrode **330_4** may be formed by being disposed and extended in the second direction **D2** from the first electrode stem **310S_4** and the second electrode stem **320S_4** and then electrically separated from the first electrode stem **310S_4** and the second electrode stem **320S_4** by a laser or the like. That is to say, the third electrode **330_4** may be substantially identical to the first electrode branch **310B_4** and the second electrode branch **320B_4**, but may be electrically separated from the electrode stems **310S_4** and **320S_4**.

(177) The display device **10_4** of FIG. **14** is different from the display device **10** of FIG. **1** in that the third electrode **330_4** may further include the electrode fragments **330Ga_4** and **330Gb_4** branching off from the first electrode stem **310S_4** and the second electrode stem **320S_4**. That is to say, the (3-1)th electrode branch **330Ba_4** and the (3-1)th electrode fragment **330Ga_4** may be formed by electrically separating the first electrode branch **310B_4** from the first electrode stem **310S_4**. The (3-2)th electrode branch **330Bb_4** and the (3-2)th electrode fragment **330Gb_4** may

be formed by electrically separating the second electrode branch **320B_4** from the second electrode stem **320S_4**.

(178) It is, however, to be understood that the present disclosure is not limited thereto. All of the third electrodes **330_4** may be formed by electrically separating the first electrode branch **310B_4** or the second electrode branch **320B_4** from the electrode stems **310S_4** and **320S_4**, respectively.

(179) The electrode branches of each of the pixels PX may be arranged in parallel in the first direction **D1**, but the ends thereof may not be aligned with one another. For example, one end of the (3-1)th electrode branch **330Ba_4** may protrude in the second direction **D2** from one end of each of the second electrode branch **320B_4** and the (3-2)th electrode branch **330Bb_4**. On the contrary, one end of the (3-2)th electrode branch **330Bb_4** may protrude in the second direction **D2** from one end of each of the first electrode branch **310B_4** and the (3-1)th electrode branch **330Ba_4**. That is to say, the plurality of electrode branches disposed in each pixel PX may include a region that is not overlapped with the region overlapping in the first direction **D1** but is partially protruded. The electrode branches of each of the pixels PX may be arranged in parallel in the first direction **D1**, but the ends thereof may not be aligned with one another. For example, one end of the (3-1)th electrode branch **330Ba_4** may protrude in the second direction **D2** from one end of each of the second electrode branch **320B_4** and the (3-2)th electrode branch **330Bb_4**. On the contrary, one end of the (3-2)th electrode branch **330Bb_4** may protrude in the second direction **D2** from one end of each of the first electrode branch **310B_4** and the (3-1)th electrode branch **330Ba_4**. That is to say, the plurality of electrode branches disposed in each pixel PX may include a region that is not overlapped with the region overlapping in the first direction **D1** but is partially protruded. It is, however, to be understood that the present disclosure is not limited thereto. When the (3-1)th electrode branch **330Ba_4** and the (3-2)th electrode branch **330Bb_4** are formed, the region cut off from the first electrode stem **310S_4** and the second electrode stem **320S_4** may be adjusted, so that the both ends of each of the electrode branches may be aligned with each other.

(180) The light-emitting elements **350** may be arranged between the first electrode branch **310B_4** and the (3-2)th electrode branch **330Bb_4**, between the (3-2)th electrode branch **330Bb_4** and the (3-1)th electrode branch **330Ba_4**, and between the (3-1)th electrode branch **330Ba_4** and the second electrode branch **320B_4**. The light-emitting elements **350** arranged between the electrode branches may be connected in series, so that they can emit light as a current flows in the first electrode branch **310B_4**, the (3-2)th electrode branch **330Bb_4**, the (3-1)th electrode branch **330Ba_4**, and the second electrode branch **320B_4**.

(181) FIGS. **15** to **17** are plan views for illustrating a method of fabricating the display device of FIG. **14**.

(182) Initially, referring to FIG. **15**, in a display device **10_4**, a plurality of electrodes, e.g., a first electrode **310_4** and a second electrode **320_4** formed on an insulating substrate layer **300**. The first electrode **310_4** and the second electrode **320_4** include electrode stems **310S_4** and **320S_4** and electrode branches **310B_4** and **320B_4**, respectively. Although FIG. **15** shows two first electrode branches **310B_4** and two second electrode branches **320B_4**, the present disclosure is not limited thereto. The arrangements and structures of the first electrode stem **310S_4**, the second electrode stem **320S_4**, the first electrode branches **310B_4** and the second electrode branches **320B_4** are identical to those described with reference to FIG. **14**. At least some of the first electrode branches **310B_4** and the second electrode branches **320B_4** may be used to form third electrode fragment **330B_4** and third electrode fragment **330G_4** via a subsequent process to be described below.

(183) Subsequently, referring to FIG. **16**, the light-emitting elements **350** are arranged between the electrode branches **310B_4** and **320B_4**. A detailed description thereon is substantially identical to that given above with reference to FIG. **9**; and, therefore, the redundant description will be omitted.

(184) Subsequently, referring to FIG. **17**, the first electrode stem **310S_4**, the second electrode stem **320S_4**, the first electrode branch **310B_4** and the second electrode branch **320B_4** are cut at the

cutting portions CB thereon. The first electrode branch **310B_4** and the second electrode branch **320B_4** electrically separated at the cutting portions CB may form the third electrode branch **330B_4** and the third electrode fragment **330G_4**. Unlike the example shown in FIG. **10**, in the example shown in FIG. **17**, the first electrode branch **310B_4** and the second electrode branch **320B_4** are also electrically disconnected at the cutting portions CB. That is to say, the third electrode **330_4** may be produced by forming a separate electrode or line, and the first electrode branch **310B_4** and the second electrode branch **320B_4** may be formed by cutting from the electrode stems **310S_4** and **320S_4**.

(185) For example, the first electrode branch **310B_4** and the second electrode branch **320B_4** located on the outer side with respect to the center of the pixel PX may be electrically connected to the first electrode stem **310S_4** and the second electrode stem **320S_4**, respectively, to receive electric signals directly from them. On the other hand, the first electrode branch **310B_4** and the second electrode branch **320B_4** located on the inner side with respect to the center of the pixel PX are electrically separated from the first electrode stem **310S_4** and the second electrode stem **320S_4**, respectively, to form the third electrode branches **330B_4**. In such a case, the third electrode fragments **330G_4** branching off from the first electrode stem **310S_4** and the second electrode stem **320S_4** may be formed together. It is, however, to be understood that the present disclosure is not limited thereto. The third electrode fragments **330G_4** may be eliminated depending on the way how the electrodes are cut.

(186) In addition, the first electrode **310_4** and the second electrode **320_4** may be electrically connected to the first thin-film transistor **120** and the power supply electrode **162** through the first electrode contact hole CNTD on the first electrode stem **310S** and the second electrode contact hole CNTS on the second electrode stem **320S**.

(187) Hereinafter, a display device according to yet another exemplary embodiment of the present disclosure will be described.

(188) FIGS. **18** and **19** are plan views of a display device according to another exemplary embodiment of the present disclosure.

(189) Referring to FIG. **18**, there may be a plurality of electrodes **330**, as described above. A display device **10_5** of FIG. **18** is identical to the display device **10_4** of FIG. **14** except that a greater number of the third electrode branches **330B_5** and a greater number of the third electrode fragments **330G_5** are disposed. In FIG. **18**, a pixel of the display device **10_5** may include six third electrode branches parts **330B_5**, and accordingly seven light-emitting elements **350** are connected in series. As the number of the aligned light-emitting elements **350** increases, the voltage-dividing efficiency of the display device and the power loss due to the line resistance can be improved, as described above. The display device **10_5** in FIG. **18** will not be described because the display device **10_4** in FIG. **14** is identical to the display device **10_2** in FIG. **11**.

(190) Referring to FIG. **19**, a display device **10_6** includes one more first electrode branch **310B_6** than the display device **10_4** of FIG. **14**, and the electrodes of the display device **10_6** have a symmetric structure with respect to the second electrode branch **320B_6**. Accordingly, each of the first light-emitting element alignment **350A_1** located on the left side of the second electrode branch **320B_6** and the second light-emitting element alignment **350A_2** located on the right side of the second electrode branch **320B_6** may include light-emitting element arrangement connected in series and may be connected to each other in parallel. A detailed description thereon is identical to that given above with reference to FIG. **12**.

(191) Incidentally, as mentioned earlier, the first electrode **310** and the second electrode **320** may be utilized as lines via which an AC power for aligning the light-emitting elements **350** is applied. In addition, a power for driving the display device **10** may be applied via the electrodes so that an electric signal may be applied to the light-emitting elements **350**. It is, however, to be understood that the present disclosure is not limited thereto. At least one of the first electrode **310** and the second electrode **320** may have different lines, one in which AC power is applied for aligning the

light emitting device **350** and another one in which the power for driving the display device **10** is applied. This will be described in detail below.

(192) FIG. **20** is a plan view of a display device according to yet another exemplary embodiment of the present disclosure.

(193) Referring to FIG. **20**, a first electrode **310_7** of a display device **10_7** includes a first electrode stem **310S_7** and a first electrode branch **310B_7** branching off from the first electrode stem **310S_7**. The first electrode stem **310S_7** and the first electrode branch **310B_7** are identical to those of the display device **10** of FIG. **1** and the display device **10_4** of FIG. **14** described above. The detailed description thereon will be omitted.

(194) A second electrode **320** may include a second electrode branch **320B'_7**, a second electrode fragment **320G'_7**, and a second electrode line **320S'_7**. Unlike the example shown in FIG. **1**, in the display device **10_7** of FIG. **20**, the second electrode branch **320B'_7** branches off from the first electrode stem **310S_7** and is electrically separated via a process to be described later. That is to say, during the process of fabricating the display device **10_7**, the first electrode branch **310B_7** is separated from the first electrode stem **310S_7**, so that the second electrode branch **320B'_7** and the second electrode fragment **320G'_7** may be formed. It is, however, to be understood that the present disclosure is not limited thereto. The second electrode branch **320B'_7** may be formed from a separate line or electrode.

(195) The second electrode branch **320B'_7** may be extended in the second direction **D2** and may be spaced apart from the first electrode branch **320B_7**. One end of the second electrode branch **320B'_7** is spaced apart from the second electrode fragment **320G'_7**, and the other end of the second electrode branch **320B'_7** may be electrically connected to the second electrode line **320S'_7** extended in the first direction **D1**.

(196) The second electrode fragment **320G'_7** may protrude from the first electrode stem **310S_7** in the second direction **D2**. The second electrode fragment **320G'_7** may be disposed to face the terminated end of the second electrode branch **320B'_7**. That is to say, the second electrode branch **320B'_7** and the second electrode fragment **320G'_7** may be aligned with each other in the second direction **D2**.

(197) The third electrode **330_7** is disposed between the first electrode branch **310B_7** and the second electrode branch **320B'_7**. The arrangement and structure of the third electrode **330_7** are identical to those described above with respect to FIGS. **1** and **14**. It is to be noted that the third electrode **330_7** may be formed by electrically separating the second electrode branch **320B_7** from the second electrode stem **320B_7** (see FIG. **21**) during the process of fabricating the display device **10_7**, as will be described later.

(198) The second electrode line **320S'_7** may be extended in the first direction **D1** and may be spaced apart from the first electrode stem **320S_7**. Although not shown in FIG. **20**, the second electrode line **320S'_7** may be disposed in different line layers from the first electrode stem **310S_7**. Accordingly, the second electrode line **320S'_7** may be electrically connected to the second electrode branch **320B'_7** through a second electrode contact hole **CNTS**. The second electrode line **320S'_7** may be connected to the power supply line **161** described above, and may apply the supply voltage to the second electrode branch **320B'_7**. On the other hand, the second electrode stem **320S_7** (see FIG. **21**) may be applied with an AC voltage for aligning the light-emitting elements **350** during the process of fabricating the display device **10_7**. That is to say, the display device **10_7** may include two different lines, i.e., the second electrode line **320S'_7** and the second electrode stem **320S_7** to thereby apply different voltages.

(199) The light-emitting elements **350** may be disposed between the first electrode branch **310B_7** and the third electrode **330_7** and between the third electrode **330_7** and the second electrode branch **320B'_7**. Some of the light-emitting elements **350** may have one ends electrically connected to the first electrode branch **310B_7** and the other ends connected to the third electrode **330_7**. The other of the light-emitting elements **350** may have one ends connected to an electrode **330_7** and

the other ends electrically connected to the second electrode branch **320B'**₇. Accordingly, the light-emitting elements **350** may be connected in series between the first electrode branch **310B**₇ and the second electrode branch **320B'**₇. The light-emitting elements **350** may receive a supply voltage from the first electrode stem **310S**₇ connected to the first thin-film transistor **120** through the first electrode contact hole CNTD and the second electrode branch **320B'**₇ connected to the power supply electrode **162** of the second electrode line **320S'**₇ through the second electrode contact hole CNTS.

(200) FIGS. **21** to **23** are plan views for illustrating a method of fabricating the display device of FIG. **20**.

(201) Referring to FIG. **21**, the first electrode stem **310S**₇, the first electrode branch **310B**₇, the second electrode stem **320S**₇, the third electrode **330**₇, the second electrode branch **320B'**₇ and the second electrode line **320S'**₇ are formed on the insulating substrate layer **300**.

(202) The first electrode stem **310S**₇ and the first electrode branch **310B**₇ are identical to those described above; and, therefore, the redundant description will be omitted.

(203) The third electrode **330**₇ branches off from the second electrode stem **320S**₇ to be extended in the second direction D2 and is separated from the first electrode stem **310S**₇. The third electrode **330**₇ is spaced apart from the first electrode branch **310B**₇, and they are extended in the opposite directions to each other in the second direction D2.

(204) The second electrode branch **320B'**₇ branches off from the first electrode stem **310S**₇ to be extended in the second direction D2 and be spaced apart from the first electrode branch **310B**₇. The second electrode branch **320B'**₇ may be extended in the same direction as the first electrode branch **310B**₇. The end of the second electrode branch **320B'**₇ may protrude more than the end of the first electrode branch **310B**₇ toward the second electrode stem **320S**₇. Because the second electrode branch **320B'**₇ is electrically connected to the second electrode line **320S'**₇ through the second electrode contact hole CNTS via a process to be described later, it may cover the second electrode line **320S'**₇ when viewed from the top.

(205) The second electrode line **320S'**₇ is extended in the first direction D1 and is disposed between the first electrode stem **310S**₇ and the second electrode stem **320S**₇. The second electrode line **320S'**₇ may be disposed in a line layer different from the first electrode stem **310S**₇ and the second electrode stem **320S**₇ and may be electrically connected through a second electrode contact hole CNTS to be described below.

(206) Subsequently, referring to FIG. **22**, one or more light-emitting elements **350** are aligned between the first electrode branch **310B**₇, the third electrode **330**₇ and the second electrode branch **320B'**₇. To align the light-emitting elements **350**, AC power is applied to the first electrode stem **310S**₇ and the second electrode stem **320S**₇ to form a capacitance induced by an electric field between the electrodes. The light-emitting elements **350** may be aligned as described above.

(207) Subsequently, referring to FIG. **23**, contact electrodes **360** are formed on the first electrode branch **310B**₇, the third electrode **330**₇ and the second electrode branch **320B**₇ to bring into contact with the light-emitting elements **350**. The above-described elements may be further formed on the contact electrodes **360**. The detailed description thereon will be omitted.

(208) Finally, as shown in FIG. **23**, the first electrode stem **310S**₇, the second electrode branch **320B'**₇, the third electrode **330**₇ and the second electrode stem **320S**₇ are electrically separated at the cutting portions CB. Then, the first electrode stem **310S**₇ may be electrically connected to the first thin-film transistor **120** through the first electrode contact hole CNTD, and the second electrode line **320S'**₇ may be electrically connected to the second electrode branch **320B'**₇ through the second electrode contact hole CNTS. In addition, the second electrode branch **320B'**₇ may be electrically separated from the first electrode stem **310S**₇ and may be spaced apart from the second electrode fragment **320G'**₇. The third electrode **330**₃ may be electrically separated from the second electrode stem **320S**₇.

(209) Accordingly, a current may flow between the first electrode branch **310B**₇ and the second

electrode branch **320B_7**. The light-emitting elements **350** disposed between the first electrode branch **310B_7** and the third electrode **330_7** and between the third electrode **330_7** and the second electrode branch **320B'_7** may be connected in series.

(210) In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications can be made to the preferred embodiments without substantially departing from the principles of the present invention. Therefore, the disclosed preferred embodiments of the invention are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A display device comprising: a first electrode stem and a second electrode line extended in a first direction and spaced from each other; a first electrode branch branching off from the first electrode stem and extended in a second direction; a second electrode branch spaced from the first electrode branch and extended in the second direction, the second electrode branch being terminated while spaced from the first electrode stem; a third electrode between the first electrode branch and the second electrode branch and extended in the second direction, both ends of the third electrode being terminated while spaced from the first electrode stem and the second electrode line; and one or more light-emitting elements located between and partially overlapping the first electrode branch and the third electrode or the second electrode branch and the third electrode, wherein the second electrode branch is electrically connected to the second electrode line.
2. A display device comprising: a first electrode stem and a second electrode line extended in a first direction and spaced from each other; a first electrode branch branching off from the first electrode stem and extended in a second direction; a second electrode branch spaced from the first electrode branch and extended in the second direction, the second electrode branch being terminated while spaced from the first electrode stem; a third electrode between the first electrode branch and the second electrode branch and extended in the second direction, both ends of the third electrode being terminated while spaced from the first electrode stem and the second electrode line; and one or more light-emitting elements located between the first electrode branch and the third electrode or the second electrode branch and the third electrode, wherein the second electrode branch is electrically connected to the second electrode line, wherein the first electrode stem and the second electrode line are in different line layers, and wherein the second electrode branch is electrically connected to the second electrode line through a contact hole.
3. The display device of claim 2, wherein the first electrode stem further comprises a second electrode fragment protruding from the first electrode stem in the second direction.
4. The display device of claim 3, wherein the second electrode fragment is extended in the second direction and is aligned with the second electrode branch.
5. The display device of claim 4, wherein the first electrode stem is electrically connected to a thin-film transistor, wherein the second electrode line is electrically connected to a common power supply line, wherein the light-emitting elements comprise a first light-emitting element between the first electrode branch and the third electrode, and a second light-emitting element between the third electrode and the second electrode branch, and wherein the first light-emitting element and the second light-emitting element are connected in series between the first electrode branch and the second electrode branch.
6. A display device comprising: a first electrode stem and a second electrode line extended in a first direction and spaced from each other, the first electrode stem and the second electrode line, in a plan view, facing each other in a second direction crossing the first direction; a first electrode branch branching off from the first electrode stem and extended in a second direction; a second electrode branch spaced from the first electrode branch and extended in the second direction, the second electrode branch being terminated while spaced from the first electrode stem; a third

electrode between the first electrode branch and the second electrode branch and extended in the second direction, both ends of the third electrode being terminated while spaced from the first electrode stem and the second electrode line; and one or more light-emitting elements located between the first electrode branch and the third electrode or the second electrode branch and the third electrode, wherein the second electrode branch is electrically connected to the second electrode line.
